TRIGONOMETRY

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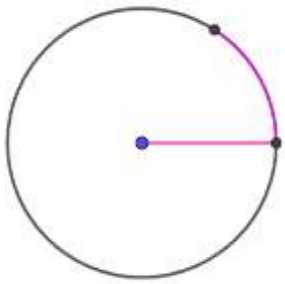
1. Equations

1.1 Basics

A. Useful Properties

1.1: Adding/Subtracting 360°

$$\sin(360°n + \theta) = \sin\theta, n \in \mathbb{Z}$$
$$\cos(360°n + \theta) = \cos\theta, n \in \mathbb{Z}$$



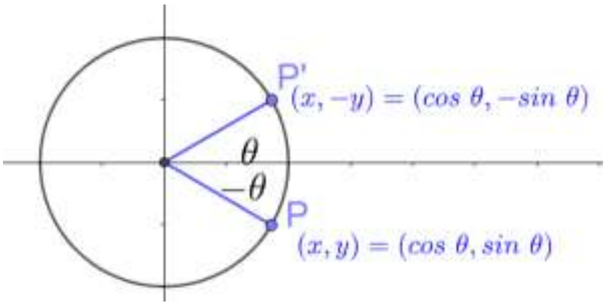
Consider an angle $x°$. The angle does not change if you:

- Add a multiple of 360°
- Subtract a multiple of 360°

1.2: Reflecting across the x –axis

$$\sin(-\theta) = -\sin\theta$$
$$\cos(-\theta) = \cos\theta$$
$$\tan(-\theta) = -\tan\theta$$

This property applies to angles in any quadrant.
But it is directly useful for converting angles in Quadrant IV into angles in Quadrant I.



$$\tan(-\theta) = \frac{\sin(-\theta)}{\cos(-\theta)} = \frac{-\sin\theta}{\cos\theta} = -\tan\theta$$

1.3: Reflecting across the y –axis

$$\sin(180° - \theta) = \sin\theta$$
$$\cos(180° - \theta) = -\cos\theta$$
$$\tan(180 - \theta) = -\tan\theta$$

This property is directly useful for converting angles in Quadrant II into angles in Quadrant I.

Let the point A have coordinates

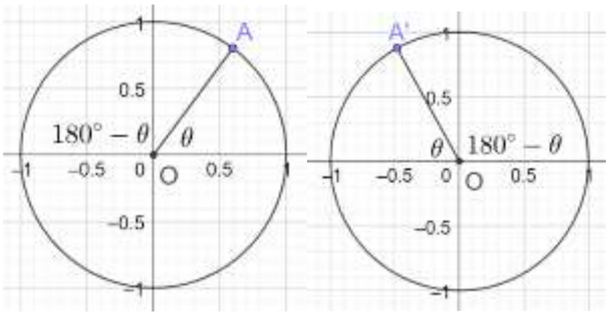
$$(x, y)$$

Let

$$\angle COB = \theta \Rightarrow \sin\theta = y, \cos\theta = x$$
$$\angle BOD = 180° - \theta \Rightarrow \sin(180 - \theta) = y$$

Hence,

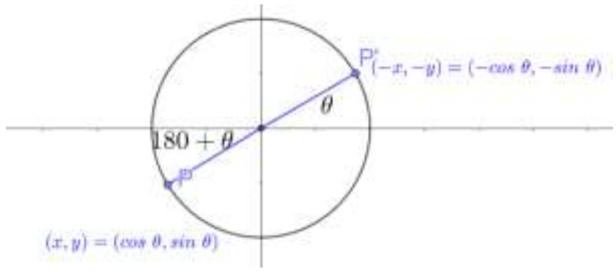
$$\sin\theta = \sin(180 - \theta)$$
$$\cos(180° - \theta) = -\cos\theta$$
$$\tan(180 - \theta) = \frac{\sin(180 - \theta)}{\cos(180 - \theta)} = \frac{\sin\theta}{-\cos\theta} = -\tan\theta$$



1.4: Reflecting across the origin

$$\sin(180° + \theta) = -\sin\theta$$
$$\cos(180° + \theta) = -\cos\theta$$
$$\tan(180° + \theta) = \tan\theta$$

This property is directly for converting angles in Quadrant III into angles in Quadrant I.



$$\tan(180° + \theta) = \frac{\sin(180° + \theta)}{\cos(180° + \theta)} = \frac{-\sin\theta}{-\cos\theta} = \frac{\sin\theta}{\cos\theta} = \tan\theta$$

B. Number of Solutions with *sin x*

Basic equations focus more on expressions involving $\sin x$ and $\cos x$. the other four trigonometric functions are also important, but occur with less frequency.

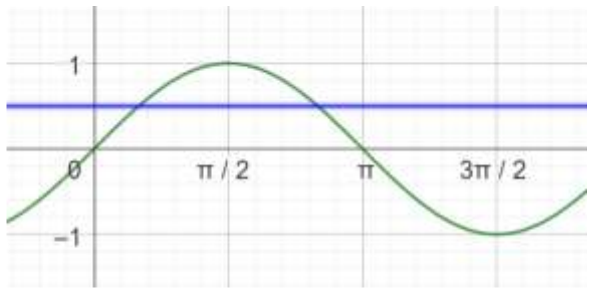
1.5: Number of Solutions

Consider the solutions of the equation below in the domain $\theta \in [0,2\pi), \theta \in \mathbb{R}$:

$$\sin\theta = c, \qquad \cos\theta = c$$

We consider three distinct cases:

- The above equation has two solutions in for $-1 < c < 1$
- It has exactly one solution for $c = \pm 1$
- If $c > 1$ or $c < -1$, then there are no solutions[1]



Example 1.6

Consider the solutions of the equation below in the domain $\theta \in [0,2\pi), \theta \in \mathbb{R}$:

$$\sin\theta = c, \qquad \cos\theta = c$$

Explain why if $c > 1$ or $c < -1$, then there are no real solutions

The range of $y = \sin x$ is

$$[-1,1]$$

If $c > 1$, the blue line $y = c$ is always above the graph of

$$y = \sin x$$

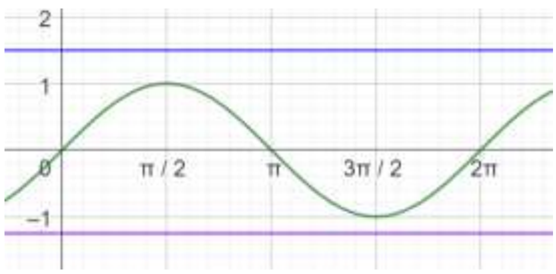
This means that it never intersects the graph.
Which means there are no solutions.

If $c < -1$, the purple line $y = c$ is always below the graph of

$$y = \sin x$$

This means that it never intersects the graph.
Which means there are no solutions.

Hence, if $c > 1$ or $c < -1$, there are no real solutions.



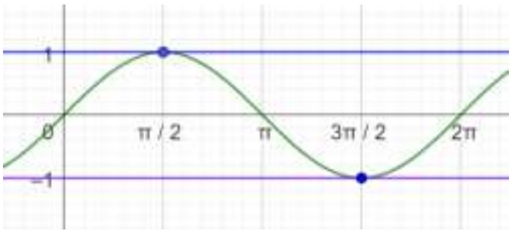
Example 1.7

Consider the solutions of the equation below in the domain $\theta \in [0,2\pi), \theta \in \mathbb{R}$:

[1] There can be complex solutions.

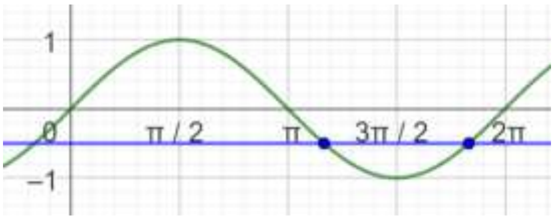
$$\sin\theta = c$$
Explain why the above equation has exactly one solution for $c = \pm 1$.

If $\sin x = 1$, the line $y = 1$ intersects the graph of $y = \sin x$ at
$$x = \frac{\pi}{2}$$
If $\sin x = -1$, the line $y = -1$ intersects the graph of $y = \sin x$ at
$$x = \frac{3\pi}{2}$$
From the graph we can see that these are the only possible solutions.



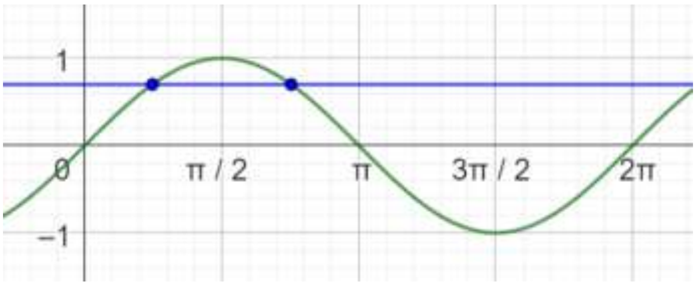
Example 1.8
Consider the solutions of the equation below in the domain $\theta \in [0,2\pi), \theta \in \mathbb{R}$:
$$\sin\theta = c$$
The above equation has two solutions for $-1 < c < 0$. Which quadrants do these lie in?

When $-1 < c < 0$, there are two solutions, shown by the places where the line intersects the graph.
One solution lies in the third quadrant, and one solution lies in the fourth quadrant.



Example 1.9
Consider the solutions of the equation below in the domain $\theta \in [0,2\pi), \theta \in \mathbb{R}$:
$$\sin\theta = c_1$$
The above equation has two solutions for $0 < c < 1$. Which quadrants do these lie in?

When $0 < c < 1$ there are two solutions, shown by the places where the line intersects the graph.
One solution lies in the first quadrant, and one solution lies in the second quadrant.



Example 1.10
Consider the following "solution" below to an equation. Without using a calculator, explain the step where the mistake is and why.
$$Step\ I\!:\ \sin x = 0.78$$
$$Step\ II\!: x \in \left\{\frac{5\pi}{4}, \frac{6\pi}{4}\right\}$$

$0 < 0.78 < 1$, and hence the solutions to it must lie in the first and the second quadrant.
However:
$$\pi < \frac{5\pi}{4} < \frac{3\pi}{2} \Rightarrow 3rd\ Quadrant \Rightarrow Invalid\ Solution$$
$$\frac{6\pi}{4} = \frac{3\pi}{2} \Rightarrow y - axis\ bottom\ value \Rightarrow Invalid\ Solution$$

Example 1.11

Without solving the equations below, identify the number of solutions to the equation, and the quadrants those solutions will lie in. If the solution does not lie in a quadrant, mention the value of x in $[0,2\pi]$.

A. $\sin x = \frac{1}{3}$
B. $\sin x = -\frac{2}{3}$
C. $\sin x = \sqrt{1}$
D. $\sin x = \sqrt{8}$
E. $\sin x = -1$
F. $\sin x = 0$

Part A

$$0 < \frac{1}{3} < 1 \Rightarrow QI, QII$$

Part B

$$-1 < -\frac{2}{3} < 0 \Rightarrow QIII, QIV$$

Part C

$$x = \frac{\pi}{2}$$

Part D

$$No\ Solutions$$

Part E

$$x = \frac{3\pi}{2}$$

Part F

$$3\ Solutions \Rightarrow x \in \{0, \pi, 2\pi\}$$

C. Equations with *sin x*

1.12: All Silver Tea Cups (Mnemonic)

In the first quadrant, ALL functions are positive.
In the second quadrant, only (SILVER) $(\sin\theta)$ is positive.
In the third quadrant, only (TEA) $\tan\theta$ is positive.
In the fourth quadrant, only (CUPS) $\cos\theta$ is positive.

Silver QII	**All** QI
Tea *QIII*	**Cups** QIV

1.13: Reflecting across the y −axis

$$\sin(180° - \theta) = \sin\theta$$

This property is directly useful for converting angles in Quadrant II into angles in Quadrant I and vice versa

Let the point A have coordinates

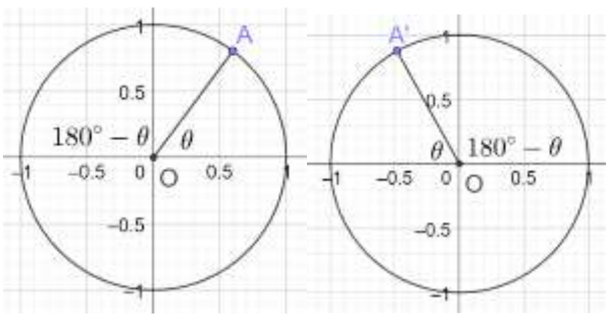
$$(x, y)$$

Let

$$\angle COB = \theta \Rightarrow \sin\theta = y, \cos\theta = x$$
$$\angle BOD = 180° - \theta \Rightarrow \sin(180 - \theta) = y$$

Hence,

$$\sin\theta = \sin(180 - \theta)$$



Example 1.14

Solve for the variable. In each case, the domain of interest is $[0,2\pi)$.

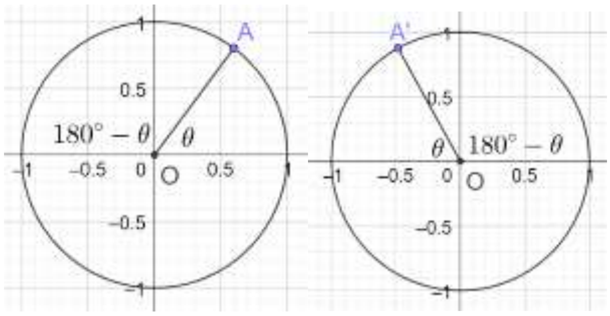
$$\sin\theta = \frac{1}{2}$$

First check where the solutions are:

$$\sin\theta = \frac{1}{2} > 0$$
$$\sin\theta > 0 \textit{ in QI and QII}$$
$$\textit{Solutions are in QI and QII.}$$

By looking at the equation, we know that a first quadrant solution is:

$$\theta_1 = \frac{\pi}{6}$$



Find the second quadrant solution

Note that $\sin\theta$ remains unchanged after the reflection across the y axis ($\sin(\pi-\theta) = \sin\theta$):

$$\theta_2 = \pi - \frac{\pi}{6} = \frac{5\pi}{6} \Rightarrow \theta \in \left\{\frac{\pi}{6}, \frac{5\pi}{6}\right\}$$

Example 1.15

Solve for the variable. In each case, the domain of interest is $[0,2\pi)$.

$$\sin\gamma = \frac{\sqrt{3}}{2}$$

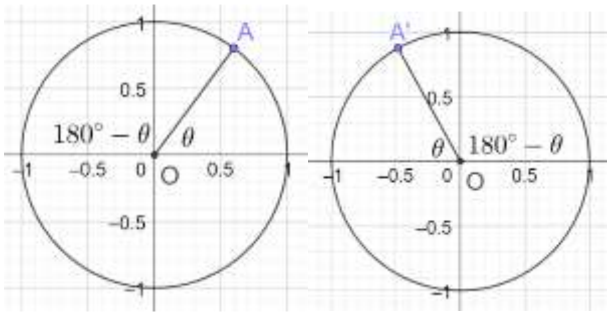
Since $\sin\gamma$ is positive, we must have solutions in

$$\textit{QI and QII}$$

First Quadrant Solution

Take the sin inverse of both sides to get the first quadrant solution:

$$\gamma_1 = \sin^{-1}\frac{\sqrt{3}}{2} = \frac{\pi}{3}$$



Find the second quadrant solution

Note that $\sin\theta$ remains unchanged after the reflection across the y axis ($\sin(\pi-\theta) = \sin\theta$):

$$\gamma_2 = \pi - \frac{\pi}{3} = \frac{2\pi}{3} \Rightarrow \gamma \in \left\{\frac{\pi}{3}, \frac{2\pi}{3}\right\}$$

Example 1.16

Solve for the variable. In each case, the domain of interest is $[0,2\pi)$.

$$\sin\phi = c, 0 < c < 1$$

Take the sin inverse of both sides to get the first quadrant solution:

$$\phi_1 = \sin^{-1} c$$

Find the second quadrant solution

Note that $\sin\theta$ remains unchanged after the reflection across the y axis ($\sin(\pi-\theta) = \sin\theta$):

$$\phi_2 = \pi - \sin^{-1} c \Rightarrow \phi \in \{\sin^{-1} c, \pi - \sin^{-1} c\}$$

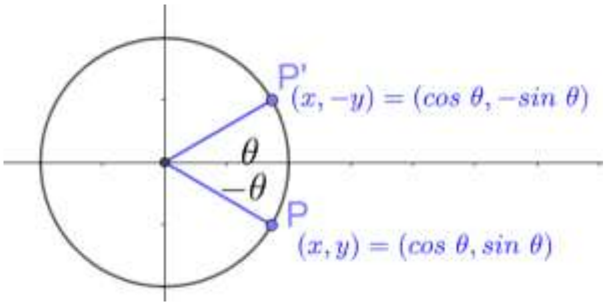
1.17: Negative Values

If you have a negative value of $\sin x$, the solutions will be in third and fourth quadrants. However, begin by finding a first quadrant solution for the equivalent positive value, and then convert it to third and fourth quadrant solutions.

1.18: Reflecting across the x −axis

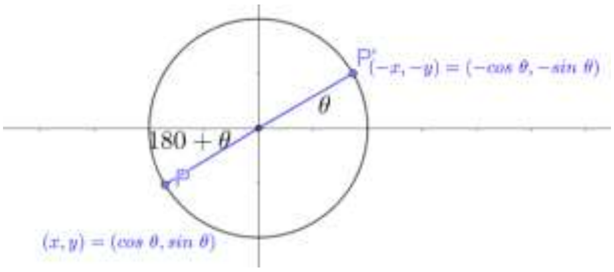
$$\sin(-\theta) = -\sin\theta$$

This property applies to angles in any quadrant.
But it is directly useful for converting angles in Quadrant IV into angles in Quadrant I.



1.19: Reflecting across the origin

$$\sin(180° + \theta) = -\sin\theta$$



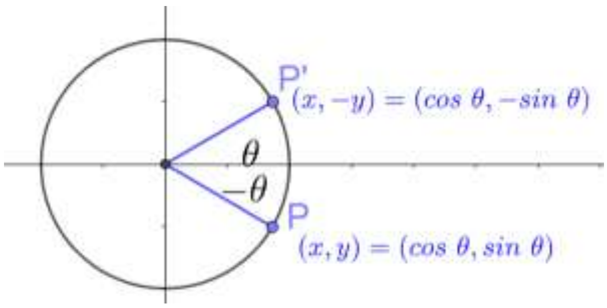
Example 1.20

Solve for the variable. In each case, the domain of interest is $[0,2\pi)$.

$$\sin\beta = -\frac{\sqrt{3}}{2}$$

sin is negative in the third and fourth quadrants. But, start by finding a first quadrant solution.

$$\sin\beta = \frac{\sqrt{3}}{2} \Rightarrow \beta = \frac{\pi}{3}$$



Reflect it across the fourth quadrant by reflecting the first quadrant solution across the x axis $(\sin(-\theta) = -\sin\theta)$:

$$\sin\frac{\pi}{3} = \frac{\sqrt{3}}{2} \Rightarrow \sin\left(-\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}$$

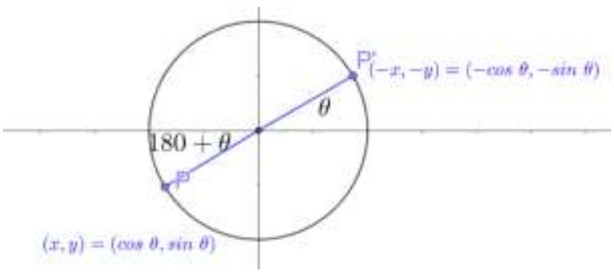
Convert the above solution to the domain of interest by using the identity $\sin(2\pi + \theta) = \sin\theta$:

$$2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$$

Use the identity $\sin(\pi + \theta) = -\sin\theta$ to find the solution in the third quadrant:

$$\sin\left(\pi + \frac{\pi}{3}\right) = \sin\frac{4\pi}{3} = -\frac{\sqrt{3}}{2}$$

$$\beta \in \left\{\frac{4\pi}{3}, \frac{5\pi}{3}\right\}$$



Example 1.21

Solve for the variable. In each case, the domain of interest is $[0,2\pi)$.

$$\sin\theta = \sqrt{2}$$

$$\sin\theta = \sqrt{2} \approx 1.41$$

But

$$-1 \le \sin\theta \le 1$$

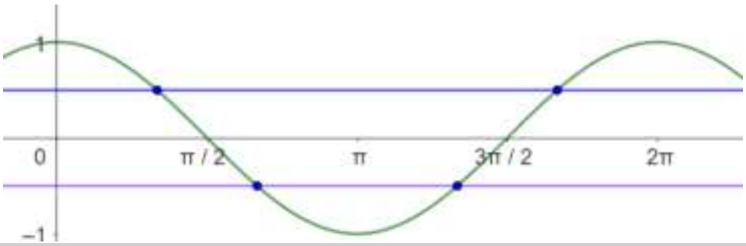
Hence, the equation has no solutions.

D. *cos x*

1.22: Number of Solutions

Consider the solutions of the equation below in the domain $\theta \in [0,2\pi), \theta \in \mathbb{R}$:

$$\cos\theta = c$$



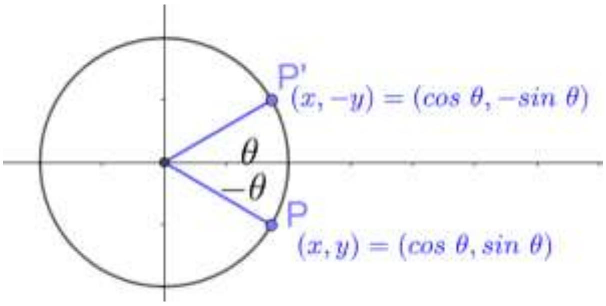
We consider four distinct cases:

- $(Blue\ Line)\ 0 < c < 1$ has two solutions (1st and 4th quadrant)
- $(Violet\ Line\ )\ -1 < c < 0$ has two solutions (2nd and 3rd quadrant)
- $c = \pm 1$ has exactly one solution
- $c > 1\ OR\ c < -1$ has no solutions[2]

Example 1.23

Solve over the domain $[0,2\pi)$:

$$\cos\alpha = \frac{1}{\sqrt{2}}$$



$$\cos\alpha > 0 \Rightarrow Solutions\ in\ QI\ and\ IV$$

$$QI\ Solution: \alpha_1 = \frac{\pi}{4}$$

To get the QIV solution, reflect the QI solution across the x axis $[\cos(-\theta) = \cos\theta]$:

$$\alpha_2 = \underbrace{-\frac{\pi}{4} = 2\pi - \frac{\pi}{4}}_{Add\ 2\pi} = \frac{7\pi}{4}$$

$$\alpha \in \left\{\frac{\pi}{4}, \frac{7\pi}{4}\right\}$$

Example 1.24

Solve over the domain $[0,2\pi)$:

$$\cos\alpha = \frac{1}{2}$$

Solutions will be first and fourth quadrant.

$$QI: 60°$$

$$QIV: -60° = -60 + 360 = 300$$

$$QI: \frac{\pi}{3}$$

[2] There can be complex solutions.

$$QIV := -\frac{\pi}{3} = -\frac{\pi}{3} + 2\pi = \frac{5\pi}{3}$$

Example 1.25

Solve over the domain $[0,2\pi)$:

$$\cos\alpha = \frac{\sqrt{3}}{2}$$

Solutions will be first and fourth quadrant.

$$QI: 30°$$
$$QIV: -30° = -30 + 360 = 330°$$

Example 1.26

Solve over the domain $[0,2\pi)$:

A. $\cos\alpha = -\frac{1}{2}$
B. $\cos\beta = -\frac{1}{\sqrt{2}}$
C. $\cos\gamma = -\frac{\sqrt{3}}{2}$

Solutions will be second and third fourth quadrant.

Part A

We first find a first quadrant solution:

$$QI: \cos\alpha = -\frac{1}{2} \Rightarrow \alpha = 60°$$

Convert the first quadrant solution to corresponding second and third quadrant solutions
Use $\cos(180-\theta) = -\cos\theta$ to get the second quadrant solution:

$$QII: 180 - 60 = 120°$$

Use $\cos(180+\theta) = -\cos\theta$ to get the third quadrant solution:

$$QII: 180 + 60 = 240°$$

Combine the solutions to get:

$$\alpha \in \{120°, 240°\}$$

Part B

$$QI: 45°$$
$$QII: 180 - 45 = 135°$$
$$QII: 180 + 45 = 225°$$
$$\alpha \in \{135°, 225°\}$$

Part C

$$QI: 30°$$
$$QII: 180 - 30 = 150°$$
$$QII: 180 + 30 = 210°$$
$$\alpha \in \{150°, 210°\}$$

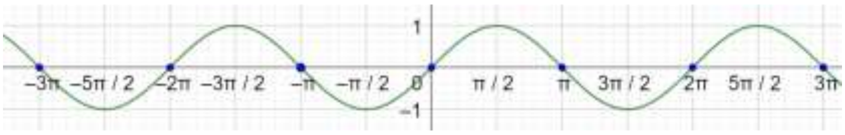
Example 1.27: Zeroes of the Trig Functions

A. Determine the zeroes of the six trigonometric functions.
B. Explain why some of the functions do not have zeroes.

sin x

From the graph of $y = \sin x$, the roots are precisely the integer multiples of π. We can write this as:

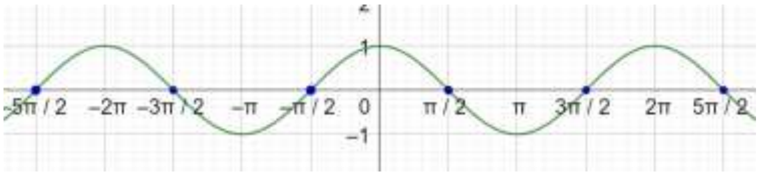
$$\sin\theta = 0 \Rightarrow \theta = n\pi, n \in \mathbb{Z}$$



cos x

Again, use the graph. The roots of $y = \cos x$ are:

$$\cos\theta = 0 \Rightarrow \theta = \frac{n\pi}{2}, \qquad n \text{ is odd integer}$$



tan x

$$\tan\theta = 0 \Rightarrow \frac{\sin\theta}{\cos\theta} = 0 \Rightarrow \sin\theta = 0 \Rightarrow \theta = n\pi, n \in \mathbb{Z}$$

$$\cot\theta = 0 \Rightarrow \theta = \frac{n\pi}{2}, \qquad n \text{ is odd integer}$$

For $\sin\theta \neq 0$:

$$\csc\theta = 0 \Rightarrow \frac{1}{\sin\theta} = 0 \Rightarrow 1 = 0 \Rightarrow \textbf{No Solutions}$$

For $\cos\theta \neq 0$:

$$\sec\theta = 0 \Rightarrow \frac{1}{\cos\theta} = 0 \Rightarrow 1 = 0 \Rightarrow \textbf{No Solutions}$$

Part B

The functions that do not have zeroes are reciprocal functions of the form:

$$\frac{1}{f(x)}$$

These do not have a zero since $\frac{1}{f(x)}$ is never zero.

E. Principal, General and Particular Solutions

1.28: Principal and General Solutions

- A solution over the domain $[0,2\pi)$ is called a principal solution.
- A solution over the domain $(-\infty,\infty)$ is called the general solution.

Example 1.29: Principal and General Solutions

Find principal and general solutions for the following equations.

A. $\sin\theta = \frac{1}{2}$

B. $\sin\phi = -\frac{\sqrt{3}}{2}$

C. $2\cos x + 1 = \frac{2\sqrt{3}+2}{2} + \cos x$

Part A

Principal Solutions

$$\sin\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$$

The above value of θ is in Quadrant I. We know that $\sin\theta$ is positive in Quadrants I and II.

Use $\sin\theta = \sin(180 - \theta)$ to the find the quadrant II solution:

$$\theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

Hence, the principal solutions are:

$$\theta = \left\{\frac{\pi}{6}, \frac{5\pi}{6}\right\}$$

General Solutions

Because the sin function is periodic with period 2π, adding 2π to any solution gives us another solution. We can generalize this as:

$$\theta = \left\{\frac{\pi}{6} + 2n\pi, \frac{5\pi}{6} + 2n\pi\right\}, n \in \mathbb{Z}$$

Part B

Principal Solutions

$$\phi = \pi + \frac{\pi}{3} = \frac{4\pi}{3}$$

The above value of θ is in Quadrant III. $\sin\theta$ is negative in Quadrants III and IV.
Use $\sin\theta = \sin(180 - \theta)$ to the find the quadrant IV solution:

$$\phi = \pi - \frac{4\pi}{3} = -\frac{\pi}{3}$$

This is in Quadrant IV. But it does not lie in the domain.
Hence, add 2π to bring it in the required domain:

$$\phi = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$$

Hence, the principal solutions are:

$$\theta = \left\{\frac{4\pi}{3}, \frac{5\pi}{3}\right\}$$

General Solutions

Because the sin function is periodic with period 2π, adding 2π to any solution gives us another solution. We can generalize this as:

$$\theta = \left\{\frac{4\pi}{3} + 2n\pi, \frac{5\pi}{3} + 2n\pi\right\}, n \in \mathbb{Z}$$

Part C

Treat $\cos x$ like a variable, and solve for it:

$$2\cos x + 1 = \frac{2\sqrt{3} + 2}{2} + \cos x$$
$$2\cos x + 1 = \sqrt{3} + 1 + \cos x$$
$$\cos x = \sqrt{3}$$

Since $\sqrt{3} > 1$, and

$$-1 \le \cos x \le 1$$

the above equation has no real solutions.

Example 1.30: Trivial Quadratics and Absolute Value Equations

Find the principal, and the general solution for the following equations:

A. $\cos^2 x = \frac{1}{4}$
B. $\sin^2 x = \frac{1}{4}$
C. $|\tan x| = 1$

Part A

$$\cos x = \pm\sqrt{\frac{1}{4}} = \pm\frac{1}{2}$$

Principal Solution:

$$\cos x = \left\{\frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}\right\}$$

General Solution:

$$\cos x = \left\{\frac{\pi}{3} + n\pi, \frac{2\pi}{3} + n\pi, n \in \mathbb{Z}\right\}$$

Part B

$$\sin x = \pm\sqrt{\frac{1}{4}} = \pm\frac{1}{2}$$

Principal Solution:

$$\sin x = \left\{\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}\right\}$$

General Solution:

$$\cos x = \left\{\frac{\pi}{6} + n\pi, \frac{5\pi}{6} + n\pi\right\}$$

Part C

$$\tan x = \pm 1$$

Case I: $\tan x = 1 \Rightarrow \sin x = \cos x$

$$x \in \left\{\frac{\pi}{4}, \frac{5\pi}{4}\right\}$$

Case II: $\tan x = -1 \Rightarrow \sin x = -\cos x$

$$x \in \left\{\frac{3\pi}{4}, \frac{7\pi}{4}\right\}$$

Combine to write the principal solutions

$$x \in \left\{\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}\right\}$$

General Solution:

$$x \in \left\{\frac{\pi}{4} + n\pi, \frac{3\pi}{4} + n\pi\right\}, n \in \mathbb{Z}$$

1.31: Particular Solutions

A solution of a trigonometric equation over a given domain is called a particular solution.

Example 1.32

Particular Solutions

Solve the equations below over the given domains.

A. $\sin\theta = \frac{\sqrt{3}}{2}, -2\pi \le \theta < \pi$

B. $\tan\theta = \frac{1}{\sqrt{3}}, -\pi \le \theta < 3\pi$

Trig Functions of Expressions

Find the general solution to the equations below:

C. $\cos(2x) = \frac{1}{2}$

D. $\cos\left(\frac{x}{3}\right) = -\frac{\sqrt{3}}{2}$

Part A

First quadrant solution is:

$$\frac{\pi}{3}$$

Second quadrant solution is:

$$\pi-\frac{\pi}{3}=\frac{2\pi}{3}$$

Subtract 2π to find the negative angles:

$$\frac{\pi}{3}-2\pi=-\frac{5\pi}{3}$$
$$\frac{2\pi}{3}-2\pi=-\frac{4\pi}{3}$$

Combine the above solutions to get the final answer:

$$\theta=\left\{-\frac{5\pi}{3},-\frac{4\pi}{3},\frac{\pi}{3},\frac{2\pi}{3}\right\}$$

Part B

First quadrant solution is:

$$\frac{\pi}{6}$$

Other solution is the third quadrant:

$$\pi+\frac{\pi}{6}=\frac{7\pi}{6}$$

Add 2π:

$$2\pi+\frac{\pi}{6}=\frac{13\pi}{6}$$
$$2\pi+\frac{7\pi}{6}=\frac{19\pi}{6}=\left(3+\frac{1}{6}\right)\pi>3\pi$$

Subtract 2π:

$$\frac{\pi}{6}-2\pi=-\frac{11\pi}{6}<-\pi$$
$$\frac{7\pi}{6}-2\pi=-\frac{5\pi}{6}$$
$$\theta=\left\{-\frac{5\pi}{6},\frac{\pi}{6},\frac{7\pi}{6},\frac{13\pi}{6}\right\}$$

Part C

Find the principal solutions first:

$$\cos 2x=\frac{1}{2}\Rightarrow 2x\in\left\{\frac{\pi}{3},\frac{5\pi}{3}\right\}\Rightarrow x\in\left\{\frac{\pi}{6},\frac{5\pi}{6}\right\}$$

Convert the principal solutions to general solutions:

$$x\in\left\{\frac{\pi}{6}+2n\pi,\frac{5\pi}{6}+2n\pi\right\}$$

Part D

$$\cos\left(\frac{x}{3}\right)=-\frac{\sqrt{3}}{2}$$

cos is negative in the second and the third quadrants.

$$\frac{x}{3}\in\left\{\frac{5\pi}{6}+2n\pi,\frac{7\pi}{6}+2n\pi\right\}$$
$$x\in\left\{\frac{5\pi}{2}+6n\pi,\frac{7\pi}{2}+6n\pi\right\}$$

1.2 Change of Domain/Variable

A. Change of Domain

Example 1.33

$$\sin x=-0.5,-\pi\le x\le 0$$

In the range $0<x<2\pi$, the solutions are:

$$x\in\{210°,330°\}$$

Subtract 360°

$$x\in\{-150°,-30°\}=\left\{-\frac{5\pi}{6},-\frac{\pi}{6}\right\}$$

Example 1.34

t lies in the interval $[0,2\pi)$

$$\cos(2t)=-\frac{\sqrt{3}}{2}$$

$$0<t<2\pi$$
$$0<2t<4\pi$$

Principal Solutions

Find the first quadrant solution:

$$2t=\frac{\pi}{6}$$

Use the identity $\cos(\pi - \theta) = -\cos\theta$ to find the second quadrant solution:

$$\pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

Use the identity $\cos(\pi + \theta) = -\cos\theta$ to find the third quadrant solution:

$$\pi + \frac{\pi}{6} = \frac{7\pi}{6}$$

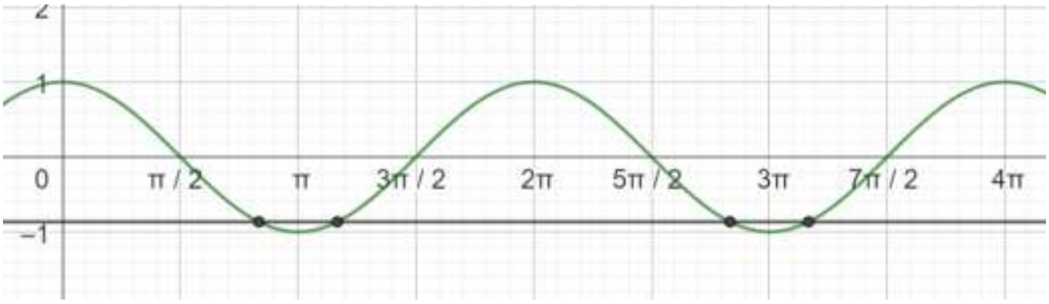
Hence, the solutions that we get in the interval $[0,2\pi)$ are these:

$$2t \in \left\{\frac{5\pi}{6}, \frac{7\pi}{6}\right\}, 2t \in [0,2\pi)$$

Domain of Interest

Find solutions in the interval $[2\pi, 4\pi)$ by adding 2π to each of the above solutions:

$$2t \in \left\{\frac{5\pi}{6}, \frac{7\pi}{6}, \frac{17\pi}{6}, \frac{19\pi}{6}\right\}, 2t \in [0,4\pi)$$



Finally, divide all the solutions by 2 to get the values for t:

$$t \in \left\{\frac{5\pi}{12}, \frac{7\pi}{12}, \frac{17\pi}{12}, \frac{19\pi}{12}\right\}, 2t \in [0,4\pi)$$

Example 1.35

Solve $\sin(2x) = -\frac{1}{2}, -\pi < x < 2\pi$

The expression inside the sin function is in terms of $2x$, while the domain is in terms of x. We need to get the domain in terms of $2x$:

$$\sin(2x) = -\frac{1}{2}, -2\pi < 2x < 4\pi$$

Find the principal solutions first:

$$2x \in \left\{\frac{7\pi}{12}, \frac{11\pi}{12}\right\}, 0 \le 2x \le 2\pi$$

Add 2π to find the solutions in the domain $2\pi \le 2x \le 4\pi$.
Subtract 2π to find the solutions in the domain $-2\pi \le 2x \le 0$.
Hence, the final solution set in terms of y:

$$2x = \left\{\frac{-17\pi}{12}, \frac{-13\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12}, \frac{31\pi}{12}, \frac{35\pi}{12}\right\}$$

$$x \in \left\{\frac{-17\pi}{24}, \frac{-13\pi}{24}, \frac{7\pi}{24}, \frac{11\pi}{24}, \frac{31\pi}{24}, \frac{35\pi}{24}\right\}$$

Example 1.36

Find the sum of the four smallest values in S given that x is in degrees and:

$$S \in \{x: -4 = 8(\sin 3(x+20)) - 8, x \in \mathbb{R}^+\}$$

Determine the domain of $3(x+20)$:

$$x > 0 \Rightarrow x + 20 > 20 \Rightarrow 3(x+20) > 60°$$

$$4 = 8(\sin 3(x+20)) \Rightarrow \frac{1}{2} = \sin 3(x+20)$$

Principal Solutions

$$3(x+20) \in \{30, 150\}$$

Determine the solutions in the domain of interest by adding 360 to the principal solutions:

$$3(x+20) \in \{150, 390, 510, 750\}$$
$$(x+20) \in \{50, 130, 170, 250\}$$
$$x \in \{30, 110, 150, 230\}$$

Example 1.37

B. Change of Variable

Example 1.38

Solve

$$\cos\left(3x + \frac{\pi}{2}\right) = \frac{\sqrt{2}}{2}, \qquad 0 \le x \le 2\pi$$

$$0 \le 3x \le 6\pi$$
$$\frac{\pi}{2} \le 3x + \frac{\pi}{2} \le \frac{13\pi}{2}$$

Let $t = 3x + \frac{\pi}{2}$

$$\cos t = \frac{\sqrt{2}}{2}, \qquad \frac{2\pi}{4} \le t \le \frac{26\pi}{4}$$

The general solutions are:

$$t \in \left\{\frac{\pi}{4}, \frac{7\pi}{4}\right\} + 2k\pi$$

$$\left\{\frac{7\pi}{4}, \frac{7\pi}{4} + 2\pi, \frac{7\pi}{4} + 4\pi\right\} = \left\{\frac{7\pi}{4}, \frac{15\pi}{4}, \frac{23\pi}{4}\right\}$$
$$\left\{\frac{9\pi}{4}, \frac{17\pi}{4}, \frac{25\pi}{4}\right\}$$

$$t = 3x + \frac{\pi}{2} \in \left\{\frac{7\pi}{4}, \frac{15\pi}{4}, \frac{23\pi}{4}, \frac{9\pi}{4}, \frac{17\pi}{4}, \frac{25\pi}{4}\right\}$$
$$3x \in \left\{\frac{5\pi}{4}, \frac{13\pi}{4}, \frac{21\pi}{4}, \frac{7\pi}{4}, \frac{15\pi}{4}, \frac{23\pi}{4}\right\}$$
$$x \in \left\{\frac{5\pi}{12}, \frac{13\pi}{12}, \frac{21\pi}{12}, \frac{7\pi}{12}, \frac{15\pi}{12}, \frac{23\pi}{12}\right\}$$

Example 1.39

$$4\cos(10x) + 2 = 2$$

$$4\cos(10x) = 0$$
$$\cos(10x) = 0$$

The principal solutions are:

$$10x \in \left\{\frac{\pi}{2}, \frac{3\pi}{2}\right\}$$

Because the period of $\cos x$ is 2π, we can add any integer multiple of 2π to the solutions to get more solutions. Hence:

$$10x \in \left\{\frac{\pi}{2}, \frac{3\pi}{2}\right\} + n \cdot 2\pi, \qquad n \in \mathbb{Z}$$

Combine the two solutions:

$$10x = \frac{\pi}{2} + n \cdot \pi, \qquad n \in \mathbb{Z}$$

Divide both sides by 10:

$$x = \frac{\pi}{20} + n \cdot \frac{\pi}{10}, \qquad n \in \mathbb{Z}$$

Example 1.40

Solve $2\cos\left(\frac{x}{2} + \frac{\pi}{2}\right) - \sqrt{3} = 0\,,\ -\pi < x < 4\pi$

Change of Variable

Let the expression inside the cosine function be equal to y:

$$\frac{x}{2} + \frac{\pi}{2} = y \Rightarrow x + \pi = 2y \Rightarrow x = 2y - \pi$$

Finding the Domain

To find the domain in terms of y, substitute the above value of y in the expression for the domain in terms of x:

$$-\pi < 2y - \pi < 4\pi$$

Add π throughout:

$$0 < 2y < 5\pi$$

Divide by 2:

$$0 < y < \frac{5}{2}\pi$$

Solve the changed equation over the changed domain

$$2\cos y - \sqrt{3} = 0 \Rightarrow \cos y = \frac{\sqrt{3}}{2} \Rightarrow y = \left\{\frac{\pi}{6}, \frac{11\pi}{6}, \frac{13\pi}{6}\right\}$$

Substitute each of the above solutions in $x = 2y - \pi$:

$$2\left\{\frac{\pi}{6}, \frac{11\pi}{6}, \frac{13\pi}{6}\right\} - \pi = \left\{-\frac{2\pi}{3}, \frac{8\pi}{3}, \frac{10\pi}{3}\right\}$$

Example 1.41: Using the original domain

$$\sin^2\left(2x - \frac{\pi}{3}\right) = \frac{3}{4}, -\pi \le x \le \pi$$

Take the square root both sides:

$$\sin\left(2x-\frac{\pi}{3}\right)=\pm\frac{\sqrt{3}}{2}$$

Take the sin inverse of both sides, and note that we get a solution in each quadrant:

$$2x-\frac{\pi}{3}\in\left\{\frac{\pi}{3},\frac{2\pi}{3},\frac{4\pi}{3},\frac{5\pi}{3}\right\}+2\pi n, n\in\mathbb{N}$$

Since $\frac{\pi}{3}+\pi=\frac{4\pi}{3},\frac{2\pi}{3}+\pi=\frac{5\pi}{3}$, we can write this as:

$$2x-\frac{\pi}{3}\in\left\{\frac{\pi}{3},\frac{2\pi}{3}\right\}+\pi n, n\in\mathbb{N}$$

Add $\frac{\pi}{3}$:

$$2x\in\left\{\frac{2\pi}{3},\pi\right\}+\pi n, n\in\mathbb{N}$$

Since the second solution requires multiples of π, substitute 0 instead of π:

$$2x\in\left\{\frac{2\pi}{3},0\right\}+\pi n, n\in\mathbb{N}$$

Divide by 2:

$$x\in\left\{0,\frac{\pi}{3}\right\}+\frac{\pi n}{2}, n\in\mathbb{N}$$

The solutions in the domain $-\pi\le x\le\pi$ are:

$$From\ x=0:\left\{-\pi,-\frac{\pi}{2},0,\frac{\pi}{2},\pi\right\}$$
$$From\ x=\frac{\pi}{3}:\left\{-\frac{2\pi}{3},-\frac{\pi}{6},\frac{\pi}{3},\frac{5\pi}{6}\right\}$$

C. Reducible to Quadratic

(Calculator) Example 1.42

$$\tan^2 x+4=2\sec^2 x+\tan x, x\in[0,2\pi]$$

Substitute $\sec^2 x=\tan^2 x+1$:

$$\tan^2 x+4=2\tan^2 x+2+\tan x$$
$$0=\tan^2 x+\tan x-2$$

Substitute $y=\tan x$:

$$y^2+y-2=0$$
$$(y+2)(y-1)=0$$
$$y\in\{-2,1\}$$

$$y=\tan x=-2\Rightarrow x=\tan^{-1}(-2)$$

Add $\pi\approx 3.14$ and $2\pi\approx 6.28$ to the above:

$$\tan^{-1}(-2)+\pi=2.03$$
$$\tan^{-1}(-2)+2\pi=5.18$$

$$y=\tan x=1\Rightarrow x=\frac{\pi}{4},\frac{5\pi}{4}$$

Example 1.43

Solve:

$$\cos x\sin^3 x-\frac{\sin x\cos x}{2}-\frac{3\sin^3 x}{4\cos x}+\frac{3\sin x}{8\cos x}=0$$

Note that $\cos x \neq 0$. Multiply both sides by $\cos x$:

$$\cos^2 x \sin^3 x - \frac{\sin x \cos^2 x}{2} - \frac{3}{4}\sin^3 x + \frac{3}{8}\sin x = 0$$

Factor out $\sin x$:

$$(\sin x)\left(\cos^2 x \sin^2 x - \frac{\cos^2 x}{2} - \frac{3}{4}\sin^2 x + \frac{3}{8}\right) = 0$$

Factor out $\cos^2 x$ from the first two terms, and $-\frac{3}{4}$ from the last two terms:

$$(\sin x)\left[(\cos^2 x)\left(\sin^2 x - \frac{1}{2}\right) - \frac{3}{4}\left(\sin^2 x - \frac{1}{2}\right)\right] = 0$$

Factor out $\sin^2 x - \frac{1}{2}$:

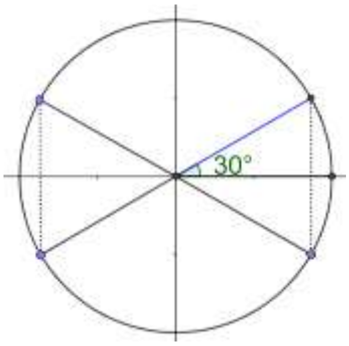
$$(\sin x)\left(\sin^2 x - \frac{1}{2}\right)\left(\cos^2 x - \frac{3}{4}\right) = 0$$

Apply the zero-product property:

$$\sin x = 0 \Rightarrow x = n\pi, \qquad n \in \mathbb{Z}$$

$$\sin^2 x - \frac{1}{2} = 0 \Rightarrow \sin x = \pm\frac{1}{\sqrt{2}} \Rightarrow x = \frac{\pi}{4} + \frac{n\pi}{2}, \qquad n \in \mathbb{Z}$$

$$\cos^2 x - \frac{3}{4} = 0 \Rightarrow \cos x = \pm\frac{\sqrt{3}}{2} \Rightarrow$$
$$x = \frac{\pi}{6} + n\pi, x = \frac{7\pi}{6} + n\pi, \qquad n \in \mathbb{Z}$$



Example 1.44

How many angles θ with $0 \le \theta \le 2\pi$ satisfy $\log(\sin(3\theta) + \log(\cos(2\theta)) = 0$? (AMC 2024 12A/8)

Use the log rule to add the terms:

$$\log[\sin(3\theta)\cos(2\theta)] = 0$$

Exponentiate both sides:

$$\sin(3\theta)\cos(2\theta) = 1$$

The sine and the cosine function both have range [-1,1]. Hence, the only solutions areL

$$\sin(3\theta) = \cos(2\theta) = 1$$
$$\sin(3\theta) = \cos(2\theta) = -1$$

But if $\sin(3\theta) = -1$, $\log(\sin(3\theta)$ is not defined.
Hence, we are left with only the positive solution.

$$0 \le \theta \le 2\pi \Rightarrow 0 \le 3\theta \le 6\pi$$

$$\sin(3\theta) = 1 \Rightarrow 3\theta \in \{90, 90 + 360, 90 + 2 \cdot 360\}$$
$$\theta \in \{30, 30 + 120, 30 + 2 \cdot 120\}$$
$$\theta \in \{30, 150, 270\}$$

$$0 \le \theta \le 2\pi \Rightarrow 0 \le 2\theta \le 4\pi$$
$$\cos(2\theta) = 1 \Rightarrow 2\theta \in \{0, 0 + 360, 0 + 2 \cdot 360\}$$
$$\theta \in \{0, 180, 360\}$$

Since the solution set of the above equations does not intersect, the number of solutions to the original equation is

$$Zero$$

1.3 Trig Identities

A. Trig Identities

1.45: Pythagorean Identity

$$\sin^2 x + \cos^2 x = 1$$
$$\cos^2 x = 1 - \sin^2 x$$
$$\sin^2 x = 1 - \cos^2 x$$

Example 1.46

Example 1.47

Find an expression for x using the tan inverse function.

$$(2\tan x + \sec x)(2\tan x - \sec x) = 1$$

$$4\tan^2 x - \sec^2 x = 1$$

Substitute $\sec^2 x = \tan^2 x + 1$:

$$4\tan^2 x - (\tan^2 x + 1) = 1$$
$$3\tan^2 x = 2$$
$$\tan x = \pm\sqrt{\frac{2}{3}}$$
$$x = \tan^{-1}\left(\pm\sqrt{\frac{2}{3}}\right)$$

Example 1.48

The number of solutions of the equation $\sin x = \cos^2 x$ in the interval $(0,10)$ is: **(JEE-M 2022)**

Substitute $\cos^2 x = 1 - \sin^2 x$:

$$\sin x = 1 - \sin^2 x$$

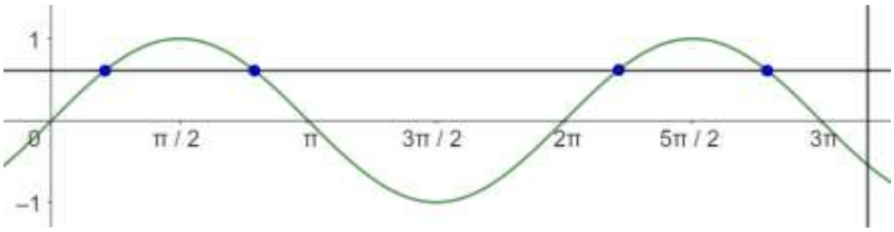
Collate all terms on the LHS to get a quadratic in $\sin x$:

$$\sin^2 x + \sin x - 1 = 0$$

Use the quadratic formula with $a = 1, b = 1, c = 1$, and substitute $\sqrt{5} \approx 2.23$:

$$\sin x = \frac{-1 \pm \sqrt{1 - (4)(1)(-1)}}{2} = \frac{-1 \pm \sqrt{5}}{2} \approx \frac{-1 \pm 2.23}{2} \in \{0.61, -1.61\}$$

Reject -1.61 since



$$-1 \le \sin x \le 1$$

$$\sin x = 0.61 \Rightarrow Valid$$

In the range:

$$[0,2\pi) \Rightarrow QI, QII \Rightarrow 2\ Solutions$$
$$[2\pi, 3\pi) \approx [6.28,9.42) \Rightarrow QI, QII \Rightarrow 2\ Solutions$$

Total number of solutions

$$= 2 + 2 = 4$$

Example 1.49

The number of values of x in the interval $\left(\frac{\pi}{4}, \frac{7\pi}{4}\right)$ for which $14\csc^2 x - 2\sin^2 x = 21 - 4\cos^2 x$, is: **(JEE-M 2022)**

$$\frac{14}{\sin^2 x} - 2\sin^2 x - 4\sin^2 x = 21 - 4\cos^2 x - 4\sin^2 x$$
$$\frac{14}{\sin^2 x} - 6\sin^2 x = 17$$

Let $t = \sin^2 x$:

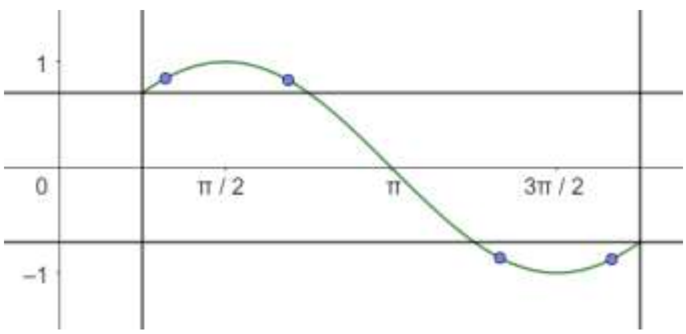
$$\frac{14}{t} - 6t = 17$$
$$14 - 6t^2 = 17t$$
$$6t^2 + 17t - 14 = 0$$
$$P = 6 \times 14 = 84 = (21)(-4)$$

$$6t^2 + 21t - 4t - 14 = 0$$
$$3t(2t + 7) - 2(2t + 7) = 0$$
$$(3t - 2)(2t + 7) = 0$$
$$\sin^2 x = t \in \left\{-\frac{7}{2}, \frac{2}{3}\right\}$$

Reject $\sin^2 x = -\frac{7}{2}$:

$$\sin^2 x = \frac{2}{3}$$

$$\sin\frac{\pi}{4} = \frac{1}{\sqrt{2}} \Rightarrow \sin^2\frac{\pi}{4} = \frac{1}{2} < \frac{2}{3}$$



$$\sin\frac{\pi}{2} = 1 \Rightarrow \sin^2\frac{\pi}{2} = 1 > \frac{2}{3}$$

Hence, there is one solution in the interval:

$$\frac{\pi}{4} < x < \frac{\pi}{2}$$

Using the similar logic, there are 3 other solutions.

$$Total: 4\ Solutions$$

1.50: Double Angle Identity for *sin*

$$\sin 2x = 2\sin x\cos x$$

Example 1.51

Find the principal solutions to

$$\sin 2x + \sin x + 2\cos x + 1 = 0$$

Substitute $\sin 2x = 2\sin x\cos x$:

$$2\sin x\cos x + \sin x + 2\cos x + 1 = 0$$

Factor:

$$\sin x\,(2\cos x + 1) + 1(2\cos x + 1) = 0$$
$$(\sin x + 1)(2\cos x + 1) = 0$$

Use the zero-product property:

$$\sin x + 1 = 0 \Rightarrow \sin x = -1 \Rightarrow x = \frac{3\pi}{2}$$
$$2\cos x + 1 = 0 \Rightarrow \cos x = -\frac{1}{2} \Rightarrow x \in \left\{\frac{2\pi}{3}, \frac{4\pi}{3}\right\}$$

$$x \in \left\{\frac{2\pi}{3}, \frac{4\pi}{3}, \frac{3\pi}{2}\right\}$$

1.52: Double Angle Identity for *cos*

$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$
$$= 2\cos^2\theta - 1$$
$$= 1 - 2\sin^2\theta$$

Example 1.53

The number of elements in the set below is: **(JEE-M 2022)**

$$S = \{\theta \in [-4\pi, 4\pi]: 3\cos^2 2\theta + 6\cos 2\theta - 10\cos^2\theta + 5 = 0\}$$

$$3\cos^2 2\theta + 6\cos 2\theta - 10\cos^2\theta + 5 = 0$$

Two of the terms have 2θ and one term has θ. Hence, we convert that term to be in terms of 2θ.

Substitute $\cos^2\theta = \frac{\cos 2\theta + 1}{2}$:

$$3\cos^2 2\theta + 6\cos 2\theta - 10\left(\frac{\cos 2\theta + 1}{2}\right) + 5 = 0$$
$$3\cos^2 2\theta + 6\cos 2\theta - 5\cos 2\theta - 5 + 5 = 0$$
$$3\cos^2 2\theta + \cos 2\theta = 0$$

Use a change of variable. Let $\cos 2\theta = x$:

$$3x^2 + x = 0 \Rightarrow x(3x+1) = 0 \Rightarrow x \in \left\{-\frac{1}{3}, 0\right\}$$

$$\theta \in [-4\pi, 4\pi] \Rightarrow 2\theta \in [-8\pi, 8\pi]$$

$$x = \cos 2\theta = 0 \Rightarrow Principal\ solutions\ are\ 2\theta \in \left\{\frac{\pi}{2}, \frac{3\pi}{2}\right\} \Rightarrow 2\ Solutions\ in\ [0,2\pi)$$

$$[-8\pi, 8\pi]$$

$$\pm(0-2\pi), \pm(2\pi-4\pi), \pm(4\pi-6\pi), \pm(6\pi-8\pi) \Rightarrow 8\ Intervals \Rightarrow 16\ Solutions$$

$$x = \cos 2\theta = -\frac{1}{3} \Rightarrow 2\ Solutions\ in\ [0,2\pi)$$

By the same logic as before:

$$\Rightarrow 16\ Solutions\ in\ [-4\pi, 4\pi]$$

Total solutions

$$16 + 16 = 32$$

1.54: Sum Formula

$$\sin\theta + \cos\theta = \sqrt{2}\sin\left(\theta + \frac{\pi}{4}\right)$$

Factor $\sqrt{2}$ in the LHS:

$$LHS = \sqrt{2}\left[(\sin\theta)\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\cos\theta\right]$$

Substitute $\sin\left(\frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$:

$$= \sqrt{2}\left[(\sin\theta)\cos\left(\frac{\pi}{4}\right) + \sin\left(\frac{\pi}{4}\right)\cos\theta\right]$$

Use the sum formula $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \sin\beta\cos\alpha$

$$= \sqrt{2}\left[\sin\left(\theta + \frac{\pi}{4}\right)\right] = RHS$$

Example 1.55

Let $S = \{\theta \in [0,2\pi): \tan(\pi\cos\theta) + \tan(\pi\sin\theta) = 0\}$. Then: **(JEE-M 2023)**

$$\sum_{\theta\in S}\sin^2\left(\theta + \frac{\pi}{4}\right) =$$

$$\tan(\pi\cos\theta) = -\tan(\pi\sin\theta)$$

Use $-\tan\theta = \tan(-\theta)$:

$$\tan(\pi\cos\theta) = \tan(-\pi\sin\theta)$$

Since the period of $\tan\theta$ is π:

$$\pi\cos\theta = n\pi - \pi\sin\theta, \qquad n \in \mathbb{Z}$$

Divide by π both sides and rearrange:

$$\sin\theta + \cos\theta = n, \qquad n \in \mathbb{Z}$$
$$\sqrt{2}\sin(\theta + 45°) = n, \qquad n \in \mathbb{Z}$$
$$\sin(\theta + 45°) = \frac{n}{\sqrt{2}}, \qquad n \in \mathbb{Z}$$

	$\sin(\theta+45°) = \frac{-1}{\sqrt{2}}$	$\sin(\theta+45°) = \frac{0}{\sqrt{2}}$	$\sin(\theta+45°) = \frac{1}{\sqrt{2}}$
	$\theta + 45° \in \{225°, 315°\}$ $\theta \in \{180°, 270°\}$	$\theta + 45° \in \{0°, 180°\}$ $\theta \in \{315°, 135°\}$	$\theta + 45° \in \{45°, 135°\}$ $\theta \in \{0°, 90°\}$
	$2\ Solutions$	$2\ Solutions$	$2\ Solutions$
	$\sin^2(\theta+45°) = \frac{1}{2}$	$\sin^2(\theta+45°) = 0$	$\sin^2(\theta+45°) = \frac{1}{2}$
$\sum_{\theta\in S}\sin^2\left(\theta + \frac{\pi}{4}\right)$	$2 \times \frac{1}{2} = 1$	$2 \times 0 = 0$	$2 \times \frac{1}{2} = 1$

Total
$$= 1 + 0 + 1 = 2$$

1.56: Sum and Difference Identities for *tan*

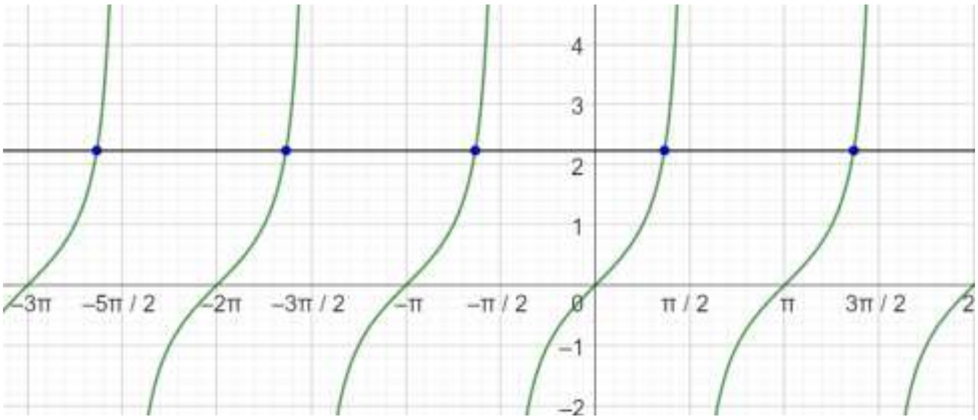
$$\tan(\alpha+\beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$$
$$\tan(\alpha-\beta) = \frac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta}$$

Example 1.57

$$S = \left[-\pi, \frac{\pi}{2}\right) - \left\{-\frac{\pi}{2}, -\frac{\pi}{4}, -\frac{3\pi}{2}\right\}$$

The number of elements in the set $A = \left\{\theta \in S : \tan\theta\left(1 + \sqrt{5}\tan(2\theta)\right) = \sqrt{5} - \tan(2\theta)\right\}$ is: **(JEE-M 2022)**

$$\tan\theta + \sqrt{5}\tan\theta\tan(2\theta) = \sqrt{5} - \tan(2\theta)$$
$$\tan\theta + \tan(2\theta) = \sqrt{5} - \sqrt{5}\tan\theta\tan(2\theta)$$
$$\frac{\tan\theta + \tan(2\theta)}{1 - \tan\theta\tan(2\theta)} = \sqrt{5}$$
$$\tan 3\theta = \sqrt{5}$$
$$3\theta = \tan^{-1}\sqrt{5} + n\pi, n \in \mathbb{N}, \qquad 3S = \left[-3\pi, \frac{3\pi}{2}\right)$$



Draw the graph (manually). There are

$$5 \textit{ solutions}$$

B. Further Questions

Example 1.58: Pythagorean Identity

Find all solutions to:

$$16^{\sin^2 x} + 16^{\cos^2 x} = 10$$

Use $\sin^2 x = 1 - \cos^2 x$

$$16^{\sin^2 x} + 16^{1-\sin^2 x} = 10 \Rightarrow 16^{\sin^2 x} + \frac{16}{16^{\sin^2 x}} = 10$$

Use a change of variable. Let $y = 16^{\sin^2 x}$:

$$y + \frac{16}{y} = 10 \Rightarrow y^2 + 16 = 10y \Rightarrow (y-8)(y-2) = 0$$

Case I: $y = 16^{\sin^2 x} = 8$

$$2^{4\sin^2 x} = 2^3 \Rightarrow 4\sin^2 x = 3 \Rightarrow \sin^2 x = \frac{3}{4} \Rightarrow \sin x = \pm\frac{\sqrt{3}}{2}$$

The solutions can be generalized to:

$$\left\{\frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}\right\} + 2n\pi, n \in \mathbb{Z} \Rightarrow \left\{\frac{\pi}{3}, \frac{2\pi}{3}\right\} + n\pi, n \in \mathbb{Z}$$

Case II: $y = 16^{\sin^2 x} = 2$

$$2^{4\sin^2 x} = 2^1 \Rightarrow 4\sin^2 x = 1 \Rightarrow \sin^2 x = \frac{1}{4} \Rightarrow \sin x = \pm\frac{1}{2}$$

Solutions are:

$$\left\{\frac{\pi}{6}, \frac{5\pi}{6}\right\} + n\pi, n \in \mathbb{Z}$$

Combine the two cases:

$$\left\{\frac{\pi}{6}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{5\pi}{6}\right\} + n\pi, n \in \mathbb{Z}$$

Example 1.59

Find the number of solutions to $9^{\sin^2 x} + 9^{\cos^2 x} = 4\sqrt{3}$ of the form $\frac{n\pi}{3}$, where n is a two-digit prime.

$$9^{\sin^2 x} + 9^{1-\sin^2 x} = 4\sqrt{3}$$
$$9^{\sin^2 x} + \frac{9}{9^{\sin^2 x}} = 4\sqrt{3}$$
$$y + \frac{9}{y} = 4\sqrt{3}$$
$$y^2 - 4\sqrt{3}y + 9 = 0$$
$$Product = 9, Sum = -4\sqrt{3} = -3\sqrt{3} - \sqrt{3}$$
$$\left(y - 3\sqrt{3}\right)\left(y - \sqrt{3}\right) = 0$$

Case I: $y = 9^{\sin^2 x} = 3\sqrt{3}$

$$9^{\sin^2 x} = 3\sqrt{3} \Rightarrow 3^{2\sin^2 x} = 3^{\frac{3}{2}} \Rightarrow 2\sin^2 x = \frac{3}{2} \Rightarrow \sin x = \pm\frac{\sqrt{3}}{2}$$

Case II: $y = 9^{\sin^2 x} = \sqrt{3}$

$$3^{2\sin^2 x} = 3^{\frac{1}{2}} \Rightarrow 2\sin^2 x = \frac{1}{2} \Rightarrow \sin^2 x = \frac{1}{4} \Rightarrow \sin x = \pm\frac{1}{2}$$

As in the previous example:

$$\left\{\frac{\pi}{6}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{5\pi}{6}\right\} + n\pi, n \in \mathbb{Z}$$

$$\left\{\frac{\pi}{3}, \frac{2\pi}{3}, \frac{3\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}, \frac{6\pi}{3}, \frac{7\pi}{3}, \dots\right\} - \left\{\frac{3\pi}{3}, \frac{6\pi}{3}, \dots\right\}$$

All values of n are available except the values where n is a multiple of 3. The only multiple of 3 which is a prime is 3 itself, and that is not valid since it is a single digit prime number.

Hence, we want the number of two-digit prime numbers which is:

$$25 - 4 = 21$$

Example 1.60: Sequences and Series

If $0 \le x \le 100\pi$, find the sum of all solutions to $\sin x + \cos x = -1$.

Simplifying the Equation

Square both sides:

$$\sin^2 x + 2\sin x\cos x + \cos^2 x = 1$$

Use the Pythagorean Identity: $\sin^2 x + \cos^2 x = 1$:

$$2\sin x\cos x + 1 = 1$$
$$2\sin x\cos x = 0$$

Use the double angle identity:

$$\sin 2x = 0$$

Finding the Principal Solution

The relevant domain for principal solutions is:

$$0 \le x < 2\pi \Rightarrow 0 \le 2x < 4\pi$$

$\sin x$ is zero at precisely integer multiples of π:

$$2x \in \{0, \pi, 2\pi, 3\pi\} \Rightarrow x \in \left\{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\right\}$$

Since we squared the equation, we need to check for extraneous solutions:

$$x \in \left\{\pi, \frac{3\pi}{2}\right\}$$

Finding the Sum $(x = \pi)$

The sum of solutions in the range $0 \le x \le 100\pi$ is:

$$\pi + 3\pi + 5\pi + \cdots + 99\pi$$
$$= \pi(1 + 3 + 5 + \cdots + 99)$$

Use the formula for the sum of the first n odd integers (99 is the 50^{th} odd integer):

$$= \pi(50^2) = 2500\pi$$

Finding the Sum $\left(x = \frac{3\pi}{2}\right)$

The sum of solutions in the range $0 \le x \le 100\pi$ is:

$$\frac{3\pi}{2} + \left(\frac{3\pi}{2} + 2\pi\right) + \left(\frac{3\pi}{2} + 4\pi\right) + \cdots + \left(\frac{3\pi}{2} + 98\pi\right)$$

Each term here is greater than the first solution by $\frac{\pi}{2}$. The final sum will be greater by:

$$50 \times \frac{\pi}{2} = 25\pi$$

The sum of solutions from this value is:

$$25\pi + 2500\pi = 2525\pi$$

Final Answer

We add the two sums:

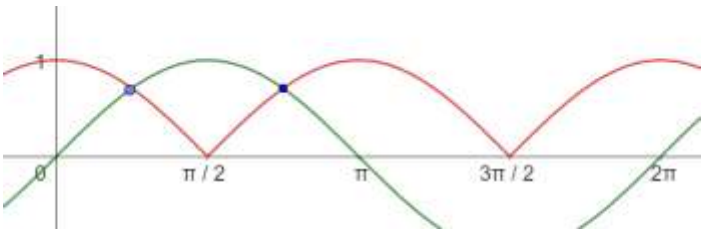
$$2500\pi + 2525\pi = 5025\pi$$

C. Graph Based Equations

Example 1.61

The number of solutions of $|\cos x| = \sin x$, such that $-4\pi \le x \le 4\pi$ is: **(JEE-M 2022)**

$$\cos x = \pm \sin x$$
$$\frac{\sin x}{\cos x} = \pm 1$$
$$\tan x = \pm 1$$



$$Principal\ Solutions: x \in \{45°, 225°\}$$
$$[0, 2\pi) \to 2\ Solutions$$
$$[0, 4\pi) \to 4\ Solutions$$
$$[-4\pi, 4\pi] \to 8\ Solutions$$

Example 1.62

Find the number of solutions to

$$\sin x = \frac{x}{8\pi}, x \ge 0$$

$$y = \sin x$$
$$y = \frac{x}{5}$$

We will get intersections only when the line is in the range of $y = \sin x$

$$For\ the\ straight\ line: -1 \le y \le 1$$
$$-1 \le \frac{x}{8\pi} \le 1$$
$$-8\pi \le x \le 8\pi$$

Since we want $x \ge 0$, we must have:

$$0 \le x \le 8\pi$$

The period for $\sin x$ is 2π. The number of cycles is:

$$\frac{8\pi}{2\pi} = 4$$

Each complete period results in two intersections.

$$No.of\ Intersections = \underbrace{4}_{\substack{No.of \\ Cycles}} \times \underbrace{2}_{\substack{Intersections \\ per\ cycle}} = 8$$

*Example 1.63

Find the number of solutions to

$$\sin x = \frac{x}{100\pi}$$

D. Identities

Example 1.64: Subjective

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$

The above is an identity. Equate the LHS and the RHS of the above identity to zero separately. Determine the principal solutions for θ. Interpret your answers.

Solve the Equation from the LHS

$$\sin 3\theta = 0$$
$$3\theta = \sin^{-1}(0) + 180k° = 0 + 180k° = 180k°$$
$$\theta = 60k°$$

Principal solutions $\big(\theta \in [0,2\pi)\big)$ are

$$\theta \in \{0, 60°, 120°, 180°, 240°, 300°\} + 360°$$

Solve the Equation from the RHS

$$3\sin\theta - 4\sin^3\theta = 0$$

Let $\alpha = \sin\theta$:

$$3\alpha - 4\alpha^3 = 0$$

$$\alpha(3-4\alpha^2)=0$$
$$\alpha=0\Rightarrow\sin\theta=0\Rightarrow\theta\in\{0,180\}$$
$$3-4\alpha^2=0\Rightarrow\alpha^2=\frac{3}{4}\Rightarrow\alpha=\pm\frac{\sqrt{3}}{2}$$
$$\sin\theta=\frac{\sqrt{3}}{2}\Rightarrow\theta\in\{60°,120°\}$$
$$\sin\theta=-\frac{\sqrt{3}}{2}\Rightarrow\theta\in\{240°,300°\}$$

Note that we obtained six solutions from the LHS, and six solutions from the RHS, and these match 1 to 1 (as they should).

Example 1.65: Subjective

$$\sin 5\theta=5\sin\theta-20\sin^3\theta+16\sin^5\theta$$

The above is an identity. Equate the LHS and the RHS of the above identity to zero separately. Determine the solutions for $-90°\le\theta\le 90°$. Interpret your answers.

Solve the LHS Equation

$$\sin 5\theta=0$$
$$5\theta=\sin^{-1}(0)+180k°=0+180k°=180k°$$
$$\theta=36k°$$

Solutions in the range $-90°\le\theta\le 90°$

$$\theta\in\{-72°,-36°,0,36°,72°\}+360°$$

Solve the RHS Equation

$$5\sin\theta-20\sin^3\theta+16\sin^5\theta=0$$

Let $\alpha=\sin\theta$:

$$5\alpha-20\alpha^3+16\alpha^5=0$$
$$\alpha(16\alpha^4-20\alpha^2+5)=0$$

If

$$\alpha=0\Rightarrow\sin\theta=0\Rightarrow\theta=0$$

If $\alpha\neq 0$, let $\beta=\alpha^2$:

$$16\beta^2-20\beta+5=0$$

Substitute $a=16, b=-20, c=5$ in the quadratic formula:

$$\alpha^2=\beta=\frac{20\pm\sqrt{400-(4)(16)(5)}}{2(16)}=\frac{5\pm\sqrt{5}}{8}\Rightarrow\alpha=\pm\sqrt{\frac{5\pm\sqrt{5}}{8}}$$

Interpreting the answers

sin 0	sin 72	sin 36	sin(−36)	sin(−72)

0	$\sqrt{\frac{5+\sqrt{5}}{8}}$	$\sqrt{\frac{5-\sqrt{5}}{8}}$	$-\sqrt{\frac{5-\sqrt{5}}{8}}$	$-\sqrt{\frac{5+\sqrt{5}}{8}}$

1.4 Logarithms

A. Logarithms

Example 1.66

$$\log_4(3\sin x)+\frac{1}{4}=\log_2\sqrt{3-\cos 2x}$$

Note that:

$$3-\cos 2x=3-(1-2\sin^2 x)=2+2\sin^2 x$$

$$\log_2\sqrt{3-\cos 2x}=\log_2(3-\cos 2x)^{\frac{1}{2}}=\frac{1}{2}\log_2(3-\cos 2x)=\log_4(3-\cos 2x)=\log_4(2+2\sin^2 x)$$

$$\frac{1}{4}=\log_4 4^{\frac{1}{4}}=\log_4\sqrt{2}$$

Substituting the above:

$$\log_4(3\sin x)+\log_4\sqrt{2}=\log_4(2+2\sin^2 x)$$

$$\log_4\left(3\sqrt{2}\sin x\right)=\log_4(2+2\sin^2 x)$$

Exponentiate both sides:

$$3\sqrt{2}\sin x=2+2\sin^2 x$$

Collate all terms on one side:

$$2\sin^2 x-3\sqrt{2}\sin x+2=0$$

Use a change of variable. Let $t=\sin x$:

$$2t-3\sqrt{2}t+2=0$$

Solve using the quadratic formula: $a=2, b=-3\sqrt{2}, c=2$

$$t=\frac{-b\pm\sqrt{b^2-4ac}}{2a}=\frac{3\sqrt{2}\pm\sqrt{18-4(2)(2)}}{4}=\frac{3\sqrt{2}\pm\sqrt{2}}{4}$$

The two solutions are:

$$t=\sin x=\frac{3\sqrt{2}+\sqrt{2}}{4}=\frac{4\sqrt{2}}{4}=\sqrt{2}>1\Rightarrow \mathit{Not\ Valid}$$

$$t=\sin x=\frac{3\sqrt{2}-\sqrt{2}}{4}=\frac{2\sqrt{2}}{4}=\frac{\sqrt{2}}{2}\Rightarrow x=\frac{\pi}{4}$$

Challenge 1.67

If $x\in[0,\pi]$ find the sum of solutions to:

$$\log_{\frac{1}{2}}\sqrt{\frac{1}{2\cos(2\pi x)+11}}=\frac{3}{2}+\log_4\left(\sqrt{3}\cos\pi x\right)$$

Use a change of variables. Substitute $y=2\cos(2\pi x)+11$ and $z=\sqrt{3}\cos\pi x$

$$\log_{\frac{1}{2}}\sqrt{\frac{1}{y}}=\frac{3}{2}+\log_4 z$$

The LHS has base $\frac{1}{2}$. The RHS has base 4. Work with the LHS to convert it to base 4.

$$LHS=\log_{\frac{1}{2}}\sqrt{\frac{1}{y}}=\log_{2^{-1}}y^{-\frac{1}{2}}=\left(-\frac{1}{2}\right)\times\left(\frac{1}{-1}\right)\log_2 y=\frac{1}{2}\log_2 y=\log_{2^2}y=\log_4 y$$

Hence, the equation becomes:

$$\log_4 y=\frac{3}{2}+\log_4 z$$

Collate all terms on the LHS, and use the Quotient Rule:

$$\log_4\frac{y}{z}=\frac{3}{2}$$

Exponentiate both sides:

$$\frac{y}{z}=4^{\frac{3}{2}}=8\Rightarrow y=8x$$

Change back to the original variables:

$$2\cos(2\pi x)+11=8\sqrt{3}\cos\pi x$$

Substitute $\cos 2\pi x=2\cos^2\pi x-1$:

$$2(2\cos^2\pi x-1)+11=8\sqrt{3}\cos\pi x$$

Expand and collate all terms on the LHS:

$$4\cos^2\pi x-8\sqrt{3}\cos\pi x+9=0$$

Use change of variables one more time. Let $\cos\pi x=t$:

$$4t^2-8\sqrt{3}t+9=0$$
$$Product=36, Sum=-8\sqrt{3}$$

$$4t-6\sqrt{3}t-2\sqrt{3}t+9=0$$
$$2t\left(2t-3\sqrt{3}\right)-\sqrt{3}\left(2t-3\sqrt{3}\right)=0$$
$$\left(2t-\sqrt{3}\right)\left(2t-3\sqrt{3}\right)=0$$

Use the zero-product property:

$$2t-3\sqrt{3}=0\Rightarrow t=\cos\pi x=\frac{3\sqrt{3}}{2}>1\Rightarrow No\ valid\ Solutions$$

$$2t-\sqrt{3}=0\Rightarrow t=\frac{\sqrt{3}}{2}\Rightarrow\cos\pi x=\frac{\sqrt{3}}{2}$$

Solutions in $[0,2\pi]$ are

$$\pi x\in\left\{\frac{\pi}{3},\frac{4\pi}{3}\right\}$$

Find the altered domain of x:

$$0 \le x \le \pi \Rightarrow 0 \le \pi x \le \pi^2$$

Since $\pi^2 \approx 9.85 > 9$,we get solutions:

$$\pi x \in \left\{\frac{\pi}{3} + 2n\pi, \frac{4\pi}{3} + 2n\pi\right\}, n = 0,1,2,3,4$$

$$x \in \left\{\frac{1}{3} + 2n, 1\frac{1}{3} + 2n\right\}, n = 0,1,2,3,4$$

$$Sum = \underbrace{\frac{1}{3} + 2\frac{1}{3} + 4\frac{1}{3} + 6\frac{1}{3} + 8\frac{1}{3}}_{First\ Quadrant} + \underbrace{1\frac{1}{3} + 3\frac{1}{3} + 5\frac{1}{3} + 7\frac{1}{3} + 9\frac{1}{3}}_{Second\ Quadrant}$$

$$1 + 2 + \cdots + 9 = \frac{9 \times 10}{2} = 45$$

$$\frac{1}{3} \times 10 = \frac{10}{3} = 3\frac{1}{3}$$

$$45 + 3\frac{1}{3} = 48\frac{1}{3}$$

1.5 Inequalities

A. Inequalities

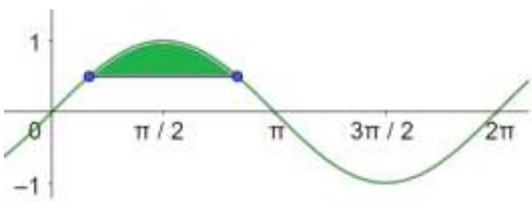
Example 1.68

Solve the following inequalities over the domain $[0,2\pi)$. Graph the related trigonometric function and shade your final answer:

A. $\sin\theta > \frac{1}{2}$
B. $\sin\theta \ge -\frac{1}{2}$
C. $\cos\theta < \frac{\sqrt{3}}{2}$
D. $\csc\theta < 2$
E. $|\csc\theta| > 2$

Part A

$\sin\theta$ is positive in the first and second quadrant:



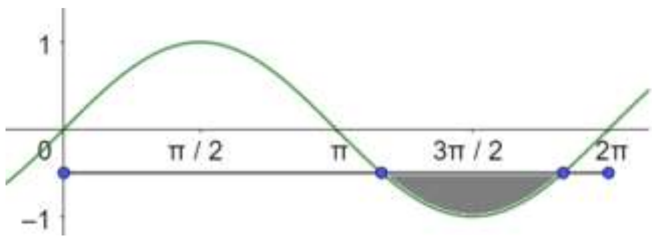
$$\sin\theta = \frac{1}{2} \Rightarrow \theta = \left\{\frac{\pi}{6}, \frac{5\pi}{6}\right\}$$

The region that we want is the shaded region between the two points.

Hence, the final answer is:

$$\sin\theta > \frac{1}{2} \Rightarrow \frac{\pi}{6} < \theta < \frac{5\pi}{6}$$

We can write the answer in interval notation as:



$$\theta \in \left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$$

Part B

$\sin\theta$ is negative in the third and fourth quadrant:

$$\sin\theta = -\frac{1}{2} \Rightarrow \theta = \left\{\frac{7\pi}{6}, \frac{11\pi}{6}\right\}$$

The region we want is above the line segment. That is, everything except the shaded region.

$$\sin\theta > -\frac{1}{2} \Rightarrow \theta \in \left[0, \frac{7\pi}{6}\right] \cup \left[\frac{11\pi}{6}, 2\pi\right)$$

Part C

$$\theta \in \left[\frac{\pi}{6}, \frac{11\pi}{6}\right]$$

Part D

$$\csc\theta < 2 \Rightarrow \frac{1}{\sin\theta} < 2 \Rightarrow \sin\theta > \frac{1}{2}$$

And this is the same inequality as Part A. Hence, it has the same solution.

Part E

Remove the absolute value sign:

$$\underbrace{\csc\theta > 2}_{Case\ I}\ OR\ \underbrace{\csc\theta < -2}_{Case\ II}$$

$$\csc\theta > 2 \Rightarrow \frac{1}{\sin\theta} > 2 \Rightarrow \sin\theta < \frac{1}{2}$$

Example 1.69

Solve the following inequalities over the domain $[0,2\pi)$. Graph the related trigonometric function and shade your final answer:

A. $\sin^2\theta \le -\frac{\sqrt{3}}{2}$
B. $\sin^2\theta \ge \frac{3}{4}$

Part A

The square of a positive real quantity cannot be negative.

$$\theta \in \phi$$

Part B

$$\sin^2\theta \ge \frac{3}{4} \Rightarrow |\sin\theta| \ge \frac{\sqrt{3}}{2}$$

Take square roots:

$$\underbrace{\sin\theta \ge \frac{\sqrt{3}}{2}}_{Case\ I} OR \underbrace{\sin\theta \le -\frac{\sqrt{3}}{2}}_{Case\ II}$$

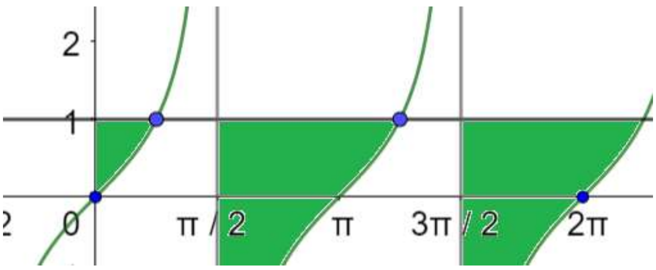
$$\theta \in \underbrace{\left[\frac{pi}{3}, \frac{2\pi}{3}\right]}_{Case\ I} \cup \underbrace{\left[\frac{4\pi}{3}, \frac{5\pi}{3}\right]}_{Case\ II}$$

Example 1.70

Solve the following inequalities over the domain $[0,2\pi)$. Graph the related trigonometric function and shade your final answer:

A. $\tan\theta < 1$
B.

Part A



$$\theta \in \left[0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{5\pi}{4}\right) \cup \left(\frac{3\pi}{2}, 2\pi\right)$$

Example 1.71

Solve the inequalities below for $x \in \mathbb{R}$:

A. $\sin x > \frac{\sqrt{3}}{2}$
B. $\csc x > 1$

C. $\cos x > \frac{1}{2}$
D. $|\sin x| \geq \frac{1}{2}$
E. $\cos^2 x \leq \frac{1}{2}$

Part A

$$\left(\frac{\pi}{3} + 2k\pi, \frac{2\pi}{3} + 2k\pi\right), k \in \mathbb{Z}$$

Part B

We consider two cases.
Case I:

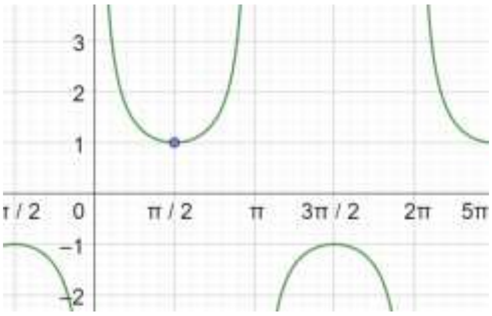
$$\sin x < 0 \Rightarrow \frac{1}{\sin x} < 1 \Rightarrow No\ Solutions$$

Case I: $\sin x > 0$

$$\csc x > 1 \Rightarrow \frac{1}{\sin x} > 1 \Rightarrow 1 > \sin x \Rightarrow \sin x < 1$$

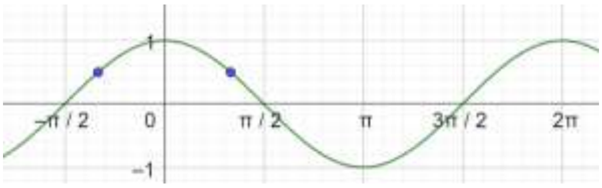
For $k \in \mathbb{Z}$:

$$(2k\pi, (2k+1)\pi) - \left\{\frac{\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \dots\right\}$$



Part C

$$\left(2k\pi - \frac{\pi}{3}, 2k\pi + \frac{\pi}{3}\right), k \in \mathbb{Z}$$



Part D

1.6 Angle Equations

A. Finding n when $n\theta = 0$

1.72: Point on the Unit Circle

A point P on the unit circle has coordinates

$$(x, y) = (\cos\theta, \sin\theta)$$

Where θ is the angle in standard position

Example 1.73

Solve $n\theta = 0$ if $\theta = \frac{\pi}{4}$ and $n \in \mathbb{N}$.

$n = 0$ is a solution but 0 is not a natural number.

Method I

$$n\theta = 0, n \in \mathbb{N}$$

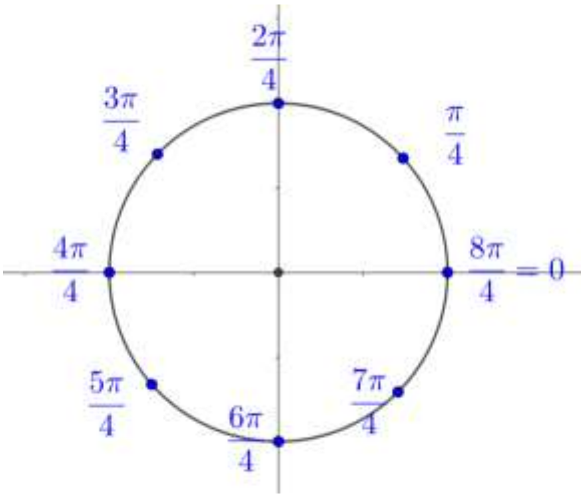
Consider the first few natural number multiples of $\frac{\pi}{4}$:

$$\left\{\frac{\pi}{4}, \frac{2\pi}{4}, \frac{3\pi}{4}, \frac{4\pi}{4}, \frac{5\pi}{4}, \frac{6\pi}{4}, \frac{7\pi}{4}, \frac{8\pi}{4}\right\}$$

$$\frac{8\pi}{4} = 2\pi = 0$$
$$n = 8 \textit{ is a solution}$$

In fact,

$$n = 8k, k \in \mathbb{N} \textit{ is a solution}$$



1.74: Adding 2π to an angle

Adding any integer multiple of 2π to an angle does not change its value.

$$\theta + 2k\pi = \theta, k \in \mathbb{Z}$$

Example 1.75

Solve $n\theta = 0$ if $\theta = \frac{\pi}{4}$ and $n \in \mathbb{N}$.

$$n\left(\frac{\pi}{4}\right) = 0$$

Since we can add 2π to any angle without changing its value, this equation should be written

$$n\left(\frac{\pi}{4}\right) = 0 + 2k\pi, k \in \mathbb{N}$$

For $k \in \mathbb{N}$:

$$n\left(\frac{\pi}{4}\right) = 2k\pi$$

Multiply both sides by $\frac{4}{\pi}$:

$$n = 2k\pi \times \frac{4}{\pi} = 8k$$

Example 1.76

Solve $n\theta = 0$ if $n \in \mathbb{N}$

A. $\theta = \frac{\pi}{3}$
B. $\theta = \frac{\pi}{8}$
C. $\theta = \frac{\pi}{2}$

Part A

$$n\left(\frac{\pi}{3}\right) = 2k\pi \Rightarrow n = 6k, k \in \mathbb{N}$$

Part B

$$n\left(\frac{\pi}{8}\right) = 2k\pi \Rightarrow n = 16k, k \in \mathbb{N}$$

Part C

$$n\left(\frac{\pi}{2}\right) = 2k\pi \Rightarrow n = 4k, k \in \mathbb{N}$$

1.77: Fractions

At a minimum k must be a natural number, but if your final answer has a denominator, then you may need to apply a stronger condition.

Example 1.78

Solve $n\theta = 0$ if $n \in \mathbb{N}$

A. $\theta = \frac{3\pi}{4}$

$$n\left(\frac{3\pi}{4}\right) = 2k\pi \Rightarrow n = \frac{8k}{3}$$

But we have to be careful here. The LHS is a natural number. Hence, the RHS must also be a natural number.

$$k = 1 \Rightarrow n = \frac{8}{3} \Rightarrow Not\ a\ natural\ number \Rightarrow Not\ Valid$$

For $\frac{8k}{3}$ to be a natural number k must be divisible by 3, and hence the final answer will be:

$$n = \frac{8k}{3}, k = 3x, x \in \mathbb{N}$$

Example 1.79

Solve $n\theta = 0$ if $n \in \mathbb{N}$

A. $\theta = \frac{5\pi}{3}$

$$n\left(\frac{5\pi}{3}\right) = 2k\pi \Rightarrow n = \frac{6k}{5}$$

Where

$$k\ is\ a\ positive\ multiple\ of\ 5$$

B. Finding θ when $n\theta = 0$

Example 1.80

Solve $n\theta = 0$ if $n = 2$ and $n \in \mathbb{N}$.

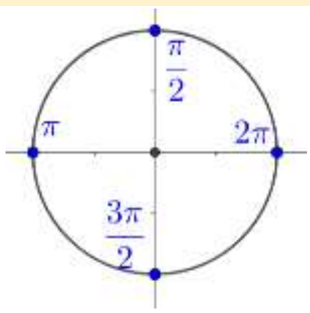
Illustrate your solutions on the unit circle.

Example 1.81

Solve $n\theta = 0$ if $n = 4$.

Illustrate your solutions on the unit circle.

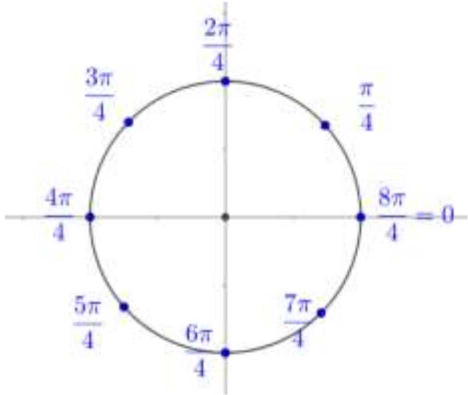
$$n\theta = 0$$
$$4\theta = 2k\pi, k \in \mathbb{Z}$$
$$\theta = \frac{k}{2}\pi, k \in \mathbb{Z}$$
$$\theta = \left\{\frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi\right\} + 2K\pi, K \in \mathbb{Z}$$



Example 1.82

Solve $n\theta = 0$ if $n = 8$.
Illustrate your solutions on the unit circle.

$$8\theta = 2k\pi \Rightarrow \theta = \frac{k}{4}\pi, k \in \mathbb{Z}$$
$$\theta = \left\{\frac{\pi}{4}, \frac{2\pi}{4}, \dots, \frac{8\pi}{4}\right\} + 2K\pi, K \in \mathbb{Z}$$
$$\theta = \left\{\frac{\pi}{4}, \frac{2\pi}{4}, \frac{3\pi}{4}, \frac{4\pi}{4}\right\} + K\pi, K \in \mathbb{Z}$$



1.83: General Solution for $\boldsymbol{\theta}$

$$n\theta = 0 \Rightarrow \theta = \frac{2k\pi}{n}, k \in \mathbb{Z}$$

$$n\theta = 0$$
$$n\theta = 2k\pi$$
$$\theta = \frac{2k\pi}{n}, k \in \mathbb{Z}$$

C. Solving general angles

Example 1.84

Solve $n\theta = \frac{\pi}{2}$ if $\theta = \frac{\pi}{4}$ and $n \in \mathbb{N}$.

$$n\theta = \frac{\pi}{2}$$
$$n\theta = \frac{\pi}{2} + 2k\pi$$
$$n\left(\frac{\pi}{4}\right) = \frac{\pi}{2} + 2k\pi$$

Multiply both sides by $\frac{4}{\pi}$:

$$n = 2 + 8k, k \in \mathbb{W}$$

1.85: Questions with no solutions

Challenge 1.86

Solve $n\theta = \frac{3\pi}{4}$ if $\theta = \frac{5\pi}{3}$ and $n \in \mathbb{N}$.

$$n\theta = \frac{3\pi}{4}$$
$$n\left(\frac{5\pi}{3}\right) = \frac{3\pi}{4} + 2k\pi$$

Multiply both sides by $\frac{12}{\pi}$:

$$20n = 9 + 24k$$
$$n = \frac{9 + 24k}{20}$$

$$n = \frac{20 + 24k - 11}{20}$$
$$n = 1 + \frac{24k - 11}{20}$$

$$\underbrace{24k}_{Even} - \underbrace{11}_{Odd} = Some\ Odd\ Number \Rightarrow Not\ divisible\ by\ 20$$
$$n \in \phi$$

D. Applications

(Non-calculator) Example 1.87

$$f(x) = \sin x^2, \qquad g(x) = \sin 2x$$

Determine the set of all solutions of the intersection of the graphs of the above two functions above for positive irrational x.

The graphs will intersect when their y values are the same.

Case I

$$f(x) = g(x)$$
$$\sin x^2 = \sin 2x$$

Since angles are unchanged by adding multiples of 2π, the underlying angles can be written:

$$x^2 = 2x + 2n\pi, \qquad n \in \mathbb{Z}$$

The above is a quadratic in x. Collate all terms on one side:

$$x^2 - 2x - 2n\pi = 0$$

Substitute $a = 1, b = -2, c = -2n\pi$ into the quadratic formula:

$$x = \frac{2 \pm \sqrt{(-2)^2 - (4)(1)(-2n\pi)}}{2(1)} = \frac{2 \pm 2\sqrt{1 + 2n\pi}}{2(1)} = 1 \pm \sqrt{1 + 2n\pi}$$

Since we want x to be positive, we cannot subtract the square root. Hence, we drop the negative sign:

$$x = 1 + \sqrt{1 + 2n\pi}, \qquad n \geq 0$$

(Since we want x to be irrational and the square root to be defined, we restrict n)

Case II

Using a unit circle identity, we can also write $\sin 2x = \sin(\pi - 2x)$:

$$\sin x^2 = \sin(\pi - 2x)$$

For integer values of n:

$$x^2 = \pi - 2x + 2n\pi$$
$$x^2 = (2n + 1)\pi - 2x$$
$$x^2 + 2x - (2n + 1)\pi = 0$$

Substitute $a = 1, b = -2, c = -2n\pi$ into the quadratic formula:

$$x = \frac{-2 \pm \sqrt{2^2 - (4)(1)(-(2n + 1)\pi)}}{2(1)} = \frac{-2 \pm 2\sqrt{1 + (2n + 1)\pi}}{2(1)} = -1 \pm \sqrt{1 + (2n + 1)\pi}$$

Since we want x to be positive, we drop the negative sign in front of the square root:

$$x = -1 + \sqrt{1 + (2n+1)\pi}, \qquad n \geq 0$$

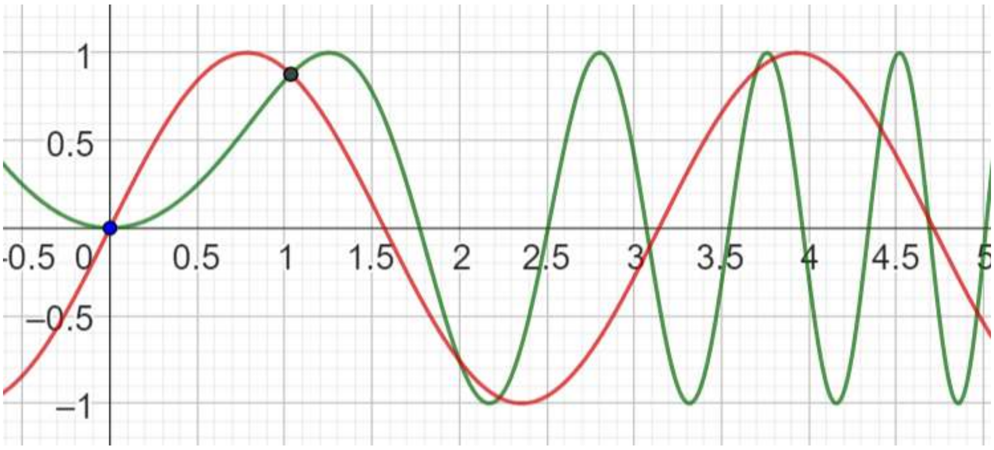
(Since we want x to be irrational and the square root to be defined, we restrict n.)

Combine Cases I and II to get:

$$x \in \left\{1 + \sqrt{1 + 2n\pi}, -1 + \sqrt{1 + (2n+1)\pi}\right\}$$

(Grapher) Example 1.88

Verify the first few solutions of the previous example by using a graphing application to graph the functions.



1.7 Composite Trigonometric Functions

A. Basics

1.89: Domain of a Composite function

$$f(g(x))$$

To determine the domain of a composite function

Step I: Determine the domain of the inner function. (Because to be a valid input, it must be in the domain of the inner function:

$$D_g = Domain\ of\ g$$

Step II: Determine the domain of the outer function

$$D_f = Domain\ of\ f$$

Step III: Remove from D_g all elements whose output is not in the domain of f.

Example 1.90

$$f(x) = \tan x\,, g(x) = \sqrt{x}$$

A. Determine the domain of $g(f(x))$.
B. Determine the domain of $f(g(x))$.

Part A

The composite function is given by:

$$g\big(f(x)\big) = \sqrt{\tan x}$$

Step I: Domain of Inner Function

$$f(x) = \tan x \Rightarrow D_{\tan x} = All\ real\ numbers\ except\ \frac{k\pi}{2}, k\ is\ odd\ integer$$

Step II: Domain of Outer Function

$$g(x) = \sqrt{x} \Rightarrow D_g\ is\ x \geq 0$$

Hence, any valid input for the outer function must be non-negative.

Step III: Ensure range of inner function is a subset of domain of Outer Function

We must that the output (range) of the inner function meets the requirements of the outer function.

$$\tan x \geq 0 \Rightarrow x \in \left[n\pi, n\pi + \frac{\pi}{2}\right), n \in \mathbb{Z}$$

$$D_{f(g(x))}\ is\ the\ set\ \left[n\pi x, n\pi x\ + \frac{\pi}{2}\right), n \in \mathbb{Z}$$

Part B

Inner Function:

$$D_{\sqrt{x}} = x \geq 0$$

Outer Function:

$$D_{\tan x} = All\ real\ numbers\ except\ \frac{k\pi}{2}, k\ is\ odd\ integer$$

Remove the values from the range of $\sqrt{x}$ which do not lie in the domain of $\tan x$:

$$\sqrt{x} = \frac{k\pi}{2} \Rightarrow x = \frac{k^2\pi^2}{4}, k\ is\ odd\ integer$$

$$D_{f(\theta)} = \left\{x \mid x \geq 0\ and\ x \neq \frac{k^2\pi^2}{4}, k\ is\ odd\ integer\right\}$$

Example 1.91

$$f(\theta) = \tan\left(\frac{\pi}{2}\cos\theta\right)$$

Determine:

A. Domain
B. Range
C. Period

Part A: Domain

Step I: Domain of inner function $\frac{\pi}{2}\cos\theta$

$$\frac{\pi}{2}\cos\theta\ has\ domain\ (-\infty, \infty)$$

Step II: Domain of outer function:

$$Domain(\tan x) \in \mathbb{R} \,/\, \frac{k\pi}{2}, k\ is\ odd\ integer$$

Step III: Ensure that the range of the inner function is a subset of the domain of the outer function

The input for the *tan* function cannot be $\frac{k\pi}{2}$. Hence, equate the inner function to the values which are not allowed:

$$\frac{\pi}{2}\cos\theta = \frac{k\pi}{2}, k\ is\ odd\ integer$$

$$\cos\theta = k, k \text{ is odd integer}$$

The only valid value of k is $k = \pm 1$:

$$\cos\theta = \pm 1 \Rightarrow \theta = n\pi, n \in \mathbb{Z}$$
$$D_{f(\theta)} = \text{All real numbers except } n\pi, n \in \mathbb{Z}$$

Part B: Range

$$g(\theta) = \frac{\pi}{2}\cos\theta$$
$$\text{Range of } g(\theta) = \frac{\pi}{2}\cos\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

If $\tan x$ has input $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, then it has range

$$(-\infty, \infty)$$

Part C: Range

Period is

$$2\pi$$

Example 1.92

$$f(\theta) = \cos(\sin\theta)$$

Determine:

A. Domain
B. Range
C. Period

Part A: Domain

$$\text{Domain of } \sin\theta : \mathbb{R}$$
$$\text{Range of } \sin\theta = [-1,1]$$

$$\text{Domain of } \cos\theta : \mathbb{R}$$

Range of inner function ($\sin\theta$) is a subset of domain of outer function ($\cos\theta$). Hence, we do not need to remove any values from the domain of $\sin\theta$.

$$\text{Range}$$

$$\text{Domain}(\tan x) \in \mathbb{R} \,/\, \frac{k\pi}{2}, k \text{ is odd integer}$$

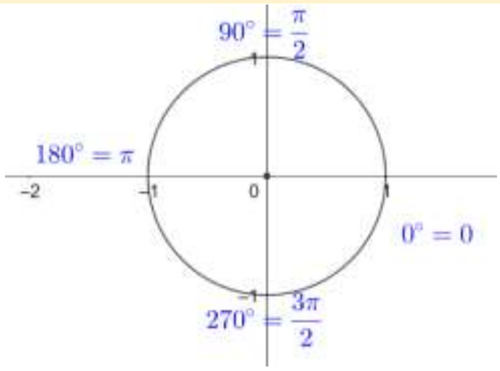
2. Graphs

2.1 Basic Graphs

A. *sin x*

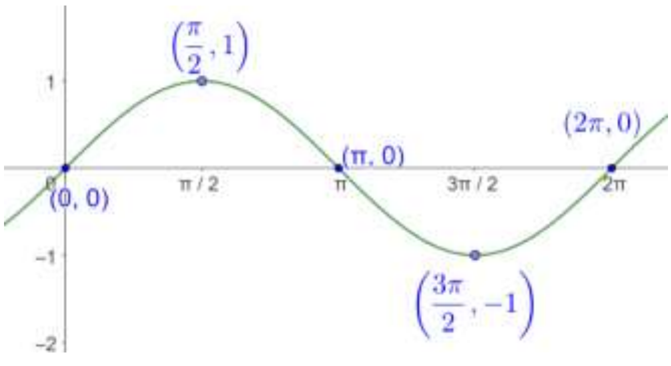
Example 2.1
From the unit circle, write:
A. $\sin 0$
B. $\sin\frac{\pi}{2}$
C. $\sin\pi$
D. $\sin\frac{3\pi}{2}$

$$\sin 0 = 0$$
$$\sin\frac{\pi}{2} = 1$$
$$\sin\pi = 0$$
$$\sin\frac{3\pi}{2} = -1$$



Example 2.2
Graph $\sin x$.

The unit circle is useful in determining the behaviour of the *sin* graph.



Example 2.3
Using the graph of $\sin x$ identify its:
A. Domain
B. Range
C. Period
D. Zeroes
E. Maxima
F. Minima

Part A

Domain is the set of all valid inputs for the function. All real numbers are valid inputs for $\sin x$. Hence, the domain is

$$x \in \mathbb{R}$$

Part B
Range is the set of valid outputs for the function. The range of $\sin x$ is:

$$[-1,1]$$

Part C
Period is where the function repeats itself. The function has a period

$$2\pi$$

Part D
Zeroes of the function are the places where the function intersects the x axis.

$$\{\dots,-2\pi,-\pi,0,\pi,2\pi,\dots\} \Rightarrow n\pi, n \in \mathbb{Z}$$

Any integer multiple of π is a zero.

Part E
The maxima are the values where $\sin x$ takes its maximum value. The maximum value is

$$y = 1$$

Which is obtained when:

$$x \in \left\{\frac{\pi}{2} + 2n\pi, n \in \mathbb{Z}\right\}$$

Part E
The minima are the values where $\sin x$ takes its minimum value. The minimum value is

$$y = -1$$

Which is obtained when:

$$x \in \left\{\frac{3\pi}{2} + 2n\pi, n \in \mathbb{Z}\right\}$$

2.4: Summary: $\sin x$

$$Domain: \mathbb{R}$$
$$Range: [-1,1]$$
$$Period: 2\pi$$
$$n\pi, n \in \mathbb{Z}$$
$$Amplitude: 1$$
$$Maxima: y = 1 \Leftrightarrow x \in \left\{\frac{\pi}{2} + 2n\pi, n \in \mathbb{Z}\right\}$$
$$Minima: y = -1 \Leftrightarrow x \in \left\{\frac{3\pi}{2} + 2n\pi, n \in \mathbb{Z}\right\}$$

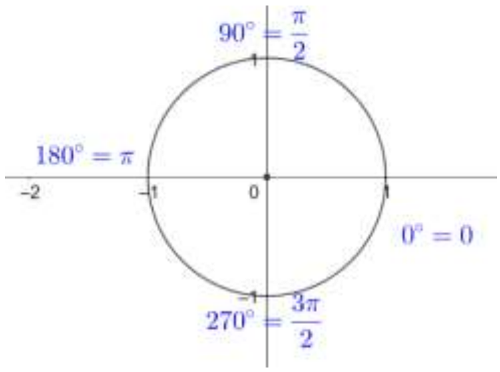
B. $\cos x$

Example 2.5
From the unit circle, write:

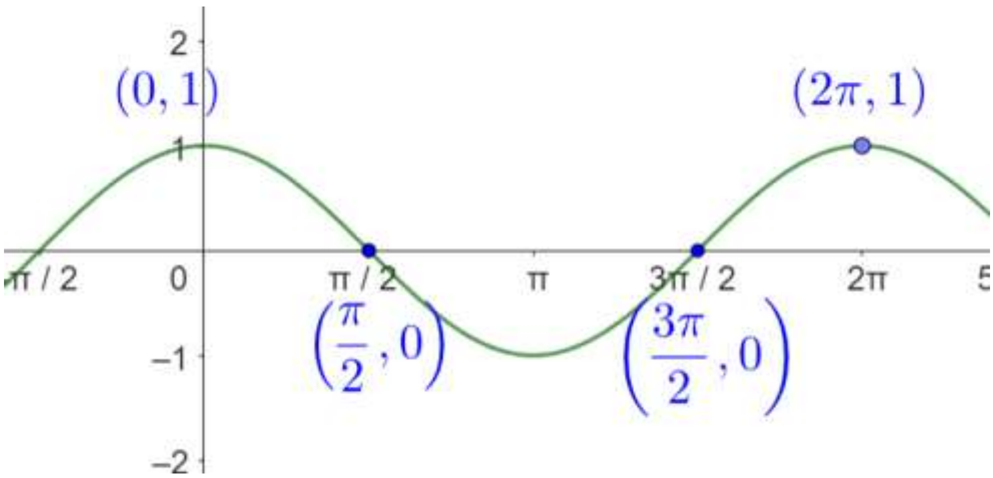
A. $\cos 0$
B. $\cos \frac{\pi}{2}$

C. $\cos\pi$
D. $\cos\frac{3\pi}{2}$

$$\cos 0 = 1$$
$$\cos\frac{\pi}{2} = 0$$
$$\cos\pi = -1$$
$$\cos\frac{3\pi}{2} = 0$$



Example 2.6
Graph $\cos x$.



2.7: $\cos x$ as a transformation of $\sin x$
The graph of $\cos x$ is obtained by shifting the graph of $\sin x$ to the left. Hence

$$\cos x = \sin\left(x+\frac{\pi}{2}\right)$$
$$\sin x = \cos\left(x-\frac{\pi}{2}\right)$$

Example 2.8
Using the graph of $\cos x$ identify its:
A. Domain
B. Range
C. Period
D. Zeroes
E. Maxima
F. Minima

Part A

Domain is the set of all valid inputs for the function. All real numbers are valid inputs for $\sin x$. Hence, the domain is

$$x \in \mathbb{R}$$

Part B

Range is the set of valid outputs for the function. The range of $\sin x$ is:

$$[-1,1]$$

Part C

Period is where the function repeats itself. The function has a period

$$2\pi$$

Part D

Zeroes of the function are the places where the function intersects the x axis.

$$\left\{\frac{\pi}{2} + n\pi, n \in \mathbb{Z}\right\}$$

Any integer multiple of π is a zero.

Part E

The maxima are the values where $\sin x$ takes its maximum value. The maximum value is

$$y = 1$$

Which is obtained when:

$$x \in 2n\pi, n \in \mathbb{Z}$$

Part E

The minima are the values where $\sin x$ takes its minimum value. The minimum value is

$$y = -1$$

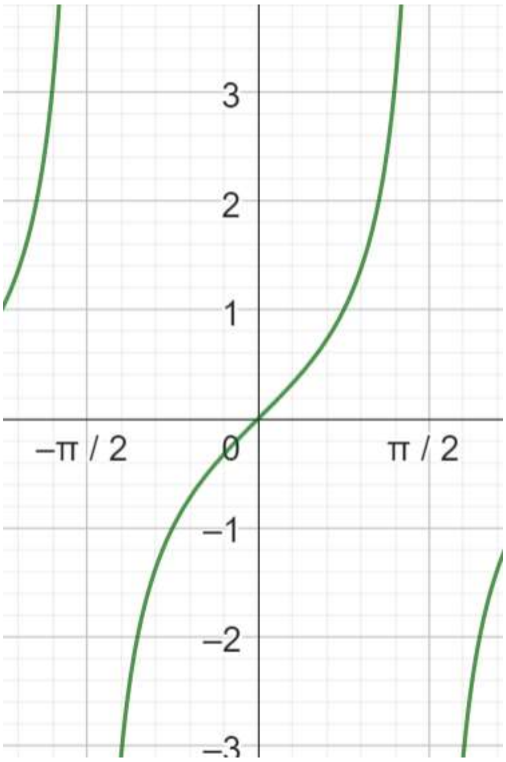
Which is obtained when:

$$x \in \{\pi + 2n\pi, n \in \mathbb{Z}\}$$

2.9: Summary of $\cos x$

$$Domain: \mathbb{R}$$
$$Range: [-1,1]$$
$$Period: 2\pi$$
$$Zeroes: \left\{\frac{\pi}{2} + n\pi, n \in \mathbb{Z}\right\}$$
$$Amplitude: 1$$
$$Maxima: y = 1 \Leftrightarrow x \in 2n\pi, n \in \mathbb{Z}$$
$$Minima: y = -1 \Leftrightarrow x \in \pi + 2n\pi, n \in \mathbb{Z}$$

C. *tan x*



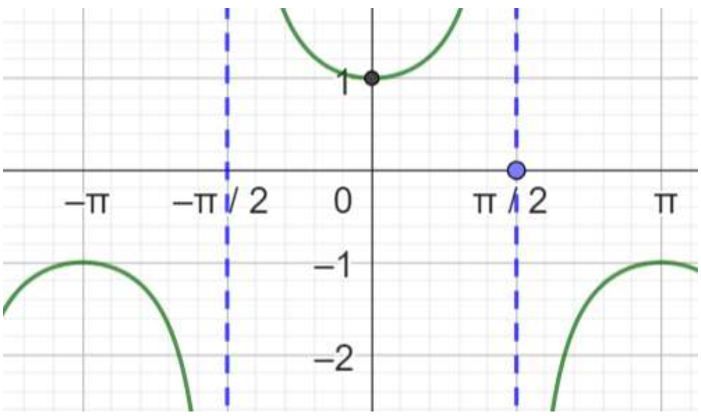
2.10: $\tan x$

$$Domain: \mathbb{R} - \left\{n\pi + \frac{\pi}{2}, \qquad n \in \mathbb{Z}\right\}$$
$$Range: \mathbb{R}$$
$$Period: \pi$$
$$Zeroes: n\pi, \qquad n \in \mathbb{Z}$$
$$Maximum: No\ maximum$$
$$Minimum: No\ minimum$$

D. Remaining Trignonometric Functions

2.11: $\sec x = \frac{1}{\cos x}$

$$Domain: \mathbb{R} - \left\{n\pi + \frac{\pi}{2}, n \in \mathbb{Z}\right\}$$
$$Range: (-\infty, -1] \cup [1, \infty)$$

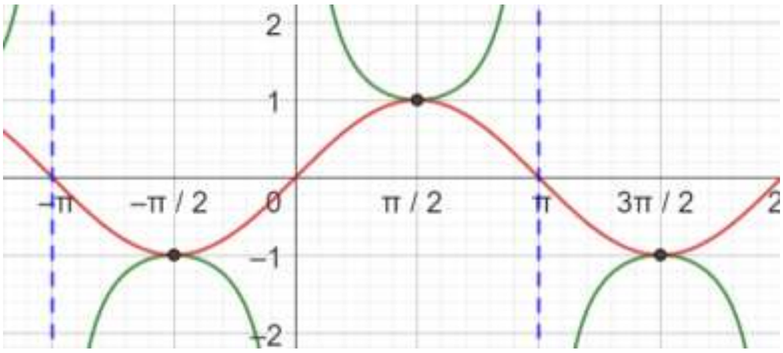


When

$$\cos x = 1 \Rightarrow \sec x = 1$$
$$\cos x = -1 \Rightarrow \sec x = -1$$

As

$$\cos x \downarrow from\ 1\ to\ 0 \Rightarrow \sec x \uparrow from\ 1\ to\ \infty$$
$$\cos x \uparrow from\ 1\ to\ 0 \Rightarrow \sec x \downarrow from -1\ to -\ \infty$$



2.12: $\csc x = \frac{1}{\sin x}$

$$Domain: \mathbb{R} - \left\{n\pi + \frac{\pi}{2}, n \in \mathbb{Z}\right\}$$
$$Range: (-\infty, -1] \cup [1, \infty)$$

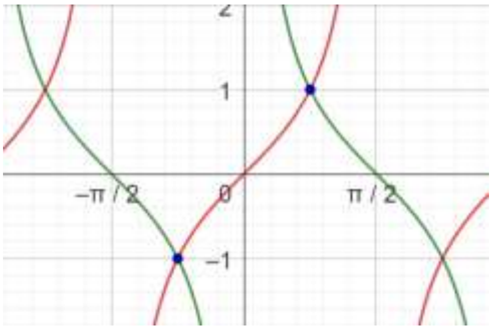
Range for $\csc x$ is the same as the range for $\sec x$

Since the graph of $\sin x$ is a shift of the graph of $\cos x$ to the right by $\frac{\pi}{2}$, therefore

$$\csc x = \frac{1}{\sin x} = \frac{1}{\cos\left(x + \frac{\pi}{2}\right)} = \sec\left(x + \frac{\pi}{2}\right)$$

2.13: $\cot x = \frac{1}{\tan x} = \frac{\cos x}{\sin x}$

$$Domain: \mathbb{R} - \left\{\frac{\pi}{2} + n\pi, n \in \mathbb{Z}\right\}$$
$$Range: (-\infty, \infty)$$



When

$$\tan x = 0 \Rightarrow \cot x \ \ has\ a\ vertical\ asymptote$$
$$\tan x \ \ is\ not\ defined \Rightarrow \cot x \ \ is\ defined\ and\ has\ value\ zero$$

E. Applications

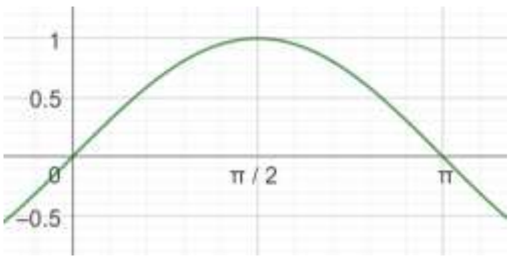
Example 2.14

Given that $0 < \theta < \frac{\pi}{7}$ rank the following in ascending order of value:

$$\sin\theta, \sin^2\theta, \sin^3\theta, \sin 2\theta, \sin 3\theta$$

The maximum of 3θ

$$= 3 \times \frac{\pi}{7} = \frac{3\pi}{7} < \frac{\pi}{2}$$



Arrange the multiples of θ

Arrange the given multiples of θ in ascending order:

$$\theta < 2\theta < 3\theta$$

For $0 < \theta < \frac{\pi}{7}$, $\sin\theta$ is an increasing function. Hence, $\theta_1 > \theta_2 \Rightarrow \sin\theta_1 > \sin\theta_2$

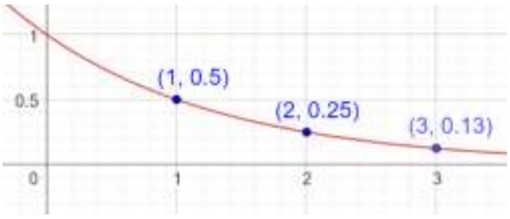
$$\underbrace{\sin\theta < \sin 2\theta < \sin 3\theta}_{Inequality\ I}$$

Arrange the powers of $\sin\theta$

We know that

$$0 < \theta < \frac{\pi}{7} \Rightarrow 0 < \sin\theta < 1$$

Let $\sin\theta = c$. For $0 < c < 1$, c^x is a decreasing function. For example, the graph alongside shows 0.5^x.



Hence, since the function is decreasing, the values will decrease as the powers increase:

$$\underbrace{\sin^3\theta < \sin^2\theta < \sin\theta}_{\textit{Inequality II}}$$

Combine the results

Combine Inequality I and II:

$$\sin^3\theta < \sin^2\theta < \sin\theta < \sin 2\theta < \sin 3\theta$$

Example 2.15

Given that θ lies in the first quadrant and n and m are natural numbers, determine the range of θ that makes the inequalities below true:

$$\sin\theta > \sin^2\theta > \cdots > \sin^{m-1}\theta > \sin^m\theta$$
$$\sin\theta < \sin 2\theta < \cdots < \sin n\theta$$

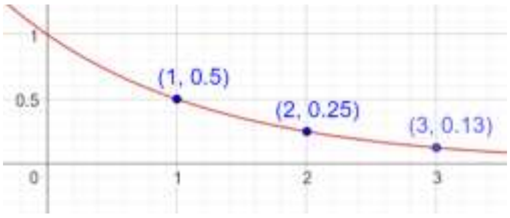
Inequality I

Since θ lies in the first quadrant,

$$0 < \sin\theta < 1$$

And increasing powers of $\sin\theta$ are successively decreasing since

$$c^x \textit{ is a decreasing function for } 0 < c < 1$$



Hence, the first inequality is always true:

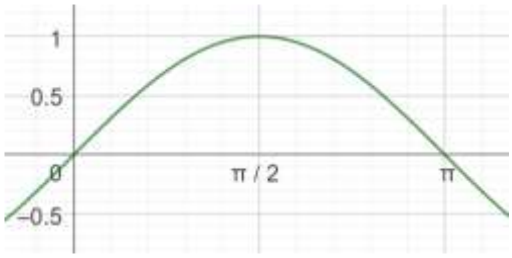
$$\sin\theta > \sin^2\theta > \cdots > \sin^{m-1}\theta > \sin^m\theta$$

Is always true.

Inequality II

$\sin\theta$ is an increasing function over

$$0 < \theta < \frac{\pi}{2}$$



To ensure that the inequality is correct, we need to ensure that the last term of the inequality has angle in the first quadrant:

$$n\theta \le \frac{\pi}{2} \Rightarrow \theta \le \frac{\pi}{2n}$$

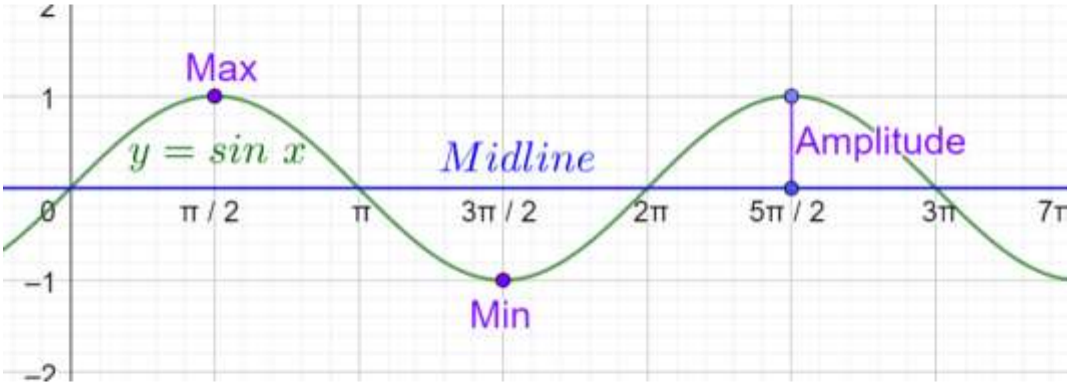
Hence, the final answer is:

$$\theta \le \frac{\pi}{2n}$$

2.2 *A* Sin(*B*x+*C*)+*D*

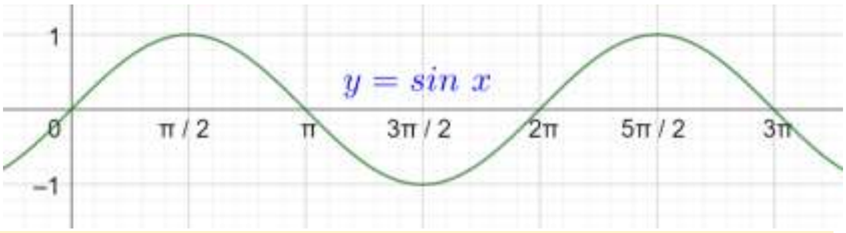
A. Basics

We start by looking at the graph of $y = \sin x$. There are many applications and extensions of what we will learn. Hence, this is an important set of concepts.



Example 2.16

The graph of $\sin x$ is drawn alongside.



A. What is the value of $\sin x$ at the origin?
B. At what other places does it take this value?
C. A periodic function is a function that repeats itself at regular intervals. What is the period of $\sin x$?

Value of $\sin x$ at the origin is 0. It is also zero at:

$$\{\dots, -2\pi, -\pi, 0\pi, \pi, 2\pi, \dots\} \Rightarrow \{0 \pm k\pi, \quad k \in \mathbb{W}\}$$
$$n\pi, \quad n \in \mathbb{Z}$$

$$Period = 2\pi$$

B. Minimum and Maximum

2.17: Peaks and Troughs

- The top point/maximum of the graphs are its peaks.
- The bottom point/minimum of the graph are its troughs.

Example 2.18

Use the graph from the previous question.

A. What is the maximum value of $\sin x$?
B. What is the smallest positive value of x for which it achieves this maximum?
C. At what other places does it achieve the maximum?
D. What is the minimum value of $\sin x$?
E. What is the smallest positive value of x for which it achieves this maximum?
F. At what other places does it achieve the maximum?

Part A

Highest value that $\sin x$ takes is 1

Part B

$$\sin\frac{\pi}{2}$$

Part C

$$\left\{\dots,-\frac{11\pi}{2},-\frac{7\pi}{2},-\frac{3\pi}{2},\frac{\pi}{2},\frac{5\pi}{2},\frac{9\pi}{2},\frac{13\pi}{2},\dots\right\}\Rightarrow\left\{\frac{\pi}{2}+2k\pi,\qquad k\in\mathbb{Z}\right\}$$

- Lowest value that $\sin x$ takes is -1
  - ✓ $\left\{\frac{3\pi}{2}\pm 2n\pi, n\in\mathbb{Z}\right\}$

Part D

$$-1$$

2.19: Midline and Amplitude

The horizontal line that goes midway between the peaks and the troughs. It is the average of the maximum and the minimum:

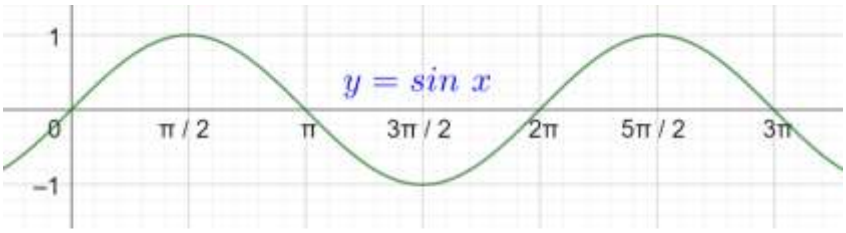
$$Midline=\frac{Max+Min}{2}$$

The distance between a peak (or a trough) and the midline is called the amplitude. The amplitude is half the distance from the maximum to the minimum:

$$Amplitude=\frac{Max-Min}{2}$$

Example 2.20

A. What is the midline of $\sin x$?
B. What is the amplitude of $\sin x$?



$$Midline=\frac{Max+Min}{2}=\frac{1-1}{2}=\frac{0}{2}=0$$
$$Amplitude=\frac{Max-Min}{2}=\frac{1-(-1)}{2}=1$$

C. $y=\sin x$ as a Function

We look at $y=\sin x$ as a function. If you have not studied functions before, look at the Note on functions before proceeding further in this section. (The rest of the chapter should still be fine if you don't know functions)

Example 2.21

Is the graph of $y=\sin x$ a function based on the vertical line test?

Any vertical line intersects the graph of $y=\sin x$ in a maximum of one place.
Hence, the graph of $y=\sin x$ is a function.

Example 2.22

Is $y = \sin x$ a function based on the algebraic test of a function?

Algebraically, we need a unique value of y for each value of x.
This is true, since for any input x, we get a unique output y.

2.23: Domain

- The domain of a function $y = f(x)$ is the set of acceptable x values for the function.
- The range of a function $y = f(x)$ is the set of y values that the function takes.

Two standard types of expressions which restrict the domain of an expression are:

- Fractions: The denominator must be non-zero, since you cannot divide by zero.
- Square Roots: In the real number system, the expression inside a square root must be positive.

Example 2.24

A. What is the domain of $f(x) = \sin x$?
B. What is the range of $f(x) = \sin x$?

There are no restrictions on the value of x. In particular, $y = \sin x$ does not have square roots or fractions. Hence, the

$$Domain\ of\ \sin x = D_f = \mathbb{R}$$

The maximum value that $\sin x$ can take is 1.

The minimum value that $\sin x$ can take is -1.

$\sin x$ is a continuous function. It takes all values between 1 and -1.

Hence, the

$$Range\ of\ \sin x = R_f = [-1,1]$$

Example 2.25: Intercepts

The x-intercepts of a function are the roots of the corresponding equation when $f(x) = 0$. The y-intercepts of a function are the point where $x = 0$. What are the intercepts of $f(x) = \sin x$?

x intercept

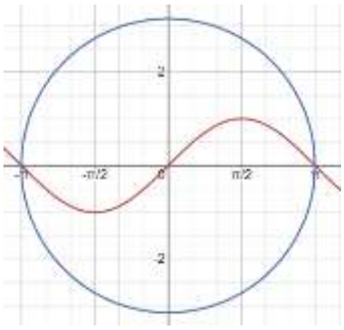
$$f(x) = 0 \Rightarrow \sin x = 0 \Rightarrow x \in \{0 \pm n\pi, \qquad n \in \mathbb{Z}\}$$

y intercept

$$x = 0 \Rightarrow \sin x = 0 \Rightarrow y\ intercept = 0$$

Example 2.26: Symmetry

A circle centered at the origin and of radius π, is divided by the curve $y = \sin x$ into two parts. The area of one of the parts is: (JMET 2011/80)



The graph of $y = \sin x$ is symmetrical about the y-axis.
Hence, the area of each part is equal.

Hence, required area:

$$= \frac{Area\ of\ Circle}{2} = \frac{\pi r^2}{2} = \frac{\pi^3}{2}$$

D. General Trigonometric Graph

2.27: General Trigonometric Graph

A basic transformation of $\sin x$ can be written in the form:

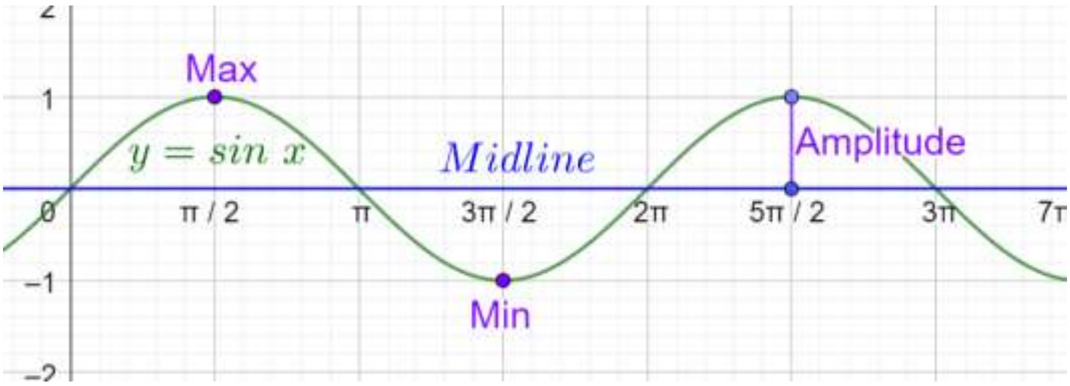
$$y = a\sin[b(x + c)] + d$$

Where

$$\underbrace{d+a}_{Max},\quad \underbrace{d-a}_{Min},\quad \underbrace{c}_{\substack{Horizontal\\Shift}},\quad \underbrace{a}_{Amplitude},\quad \underbrace{d}_{\substack{Midline\\Vertical\ Shift}},\quad \underbrace{\frac{2\pi}{b}}_{\substack{Horizontal\ Scaling\\OR\ Period}}$$

$$Max = Midline + Amplitude = d + a$$
$$Min = Midline - Amplitude = d - a$$

$$Midline = \frac{Max + Min}{2}$$
$$Amplitude = \frac{Max - Min}{2}$$



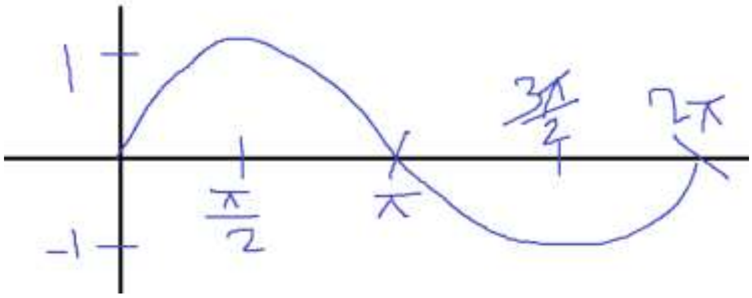
2.28: Graph Sketching Instructions

For all trigonometric graphs, mark

A. Maximum/Crest
B. Minimum/Trough
C. Midline
D. Amplitude
E. y intercept
F. x intercept

Example 2.29

Sketch $y = \sin x$ on plain paper.



E. Vertical Scaling/Amplitude: a

2.30: Vertical Scaling

- The graph of $kf(x)$ is the graph of $f(x)$ scaled vertically by a factor of k.
- In a trigonometric graph $y = a\sin[b(x+c)] + d$, the scaling factor is given by a
- The scaling factor (for a trigonometric graph) is called the amplitude.

Example 2.31

Identify the scaling factor in the following functions compared to $f(x)$:

A. $2f(x)$
B. $\frac{f(x)}{\pi}$

$$Vertically\ scaled\ by\ a\ factor\ of\ 2$$
$$Vertically\ scaled\ by\ a\ factor\ of\ \frac{1}{\pi}\ OR\ Vertically\ shrunk\ by\ a\ factor\ of \pi$$

2.32: Amplitude

$$y = a\sin x$$
Has amplitude
$$a$$

$$y = \sin x \Rightarrow Max = 1, Min = -1$$
$y = a\sin x$ scales $y = \sin x$ vertically by a factor of a:
$$Max = a, Min = -a$$
$$Amplitude = \frac{Max - Min}{2} = \frac{a-(-a)}{2} = \frac{2a}{a} = a$$

Example 2.33

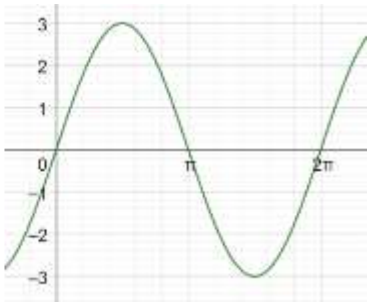
Identify the amplitude of the following, and then sketch the graph:

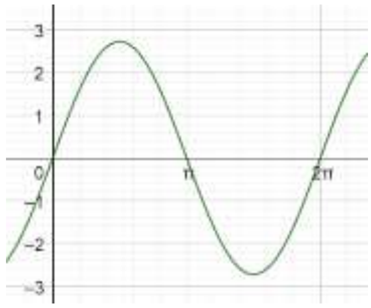
A. $y = 3\sin x$
B. $y = e\sin x$
C. $y = \pi\sin x$
D. $y = \frac{1}{e}\sin x$

Parts A and B

3

e

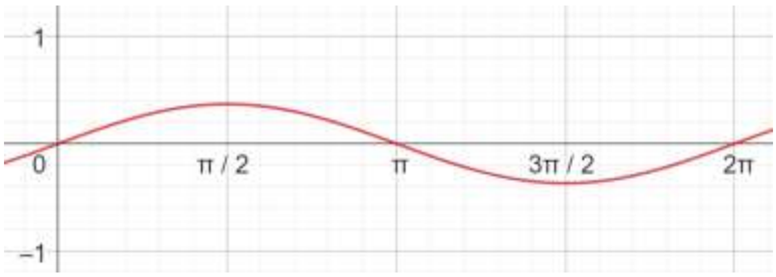




Part C

π

Part D



F. Horizontal Scaling/Period: *b*

(Important) 2.34: Horizontal Scaling

The graph of $f(kx)$ is the graph of $f(x)$ scaled horizontally by a factor of $\frac{1}{k}$.

Note that the function has kx, and the scaling is by a factor of $\frac{1}{k}$.
In other words, the scaling is the reciprocal of the factor in the function.

2.35: Period

The period of $\sin bx$ is:

$$Period = \frac{2\pi}{\underbrace{b}_{Period}}$$

The period for $y = \sin x$ is 2π.
b is a scaling factor. And since it is a horizontal scaling factor, if the coefficient of x in the function is b, then the period is scaled by its reciprocal

$$\frac{1}{b}$$

Example 2.36

Identify the period in the following, and then sketch the graph.

A. $y = \sin 2x$
B. $y = \sin \frac{1}{3}x$
C. $y = \sin \pi x$
D. $y = \sin ex$
E. $y = \sin \frac{x}{\pi}$

Parts A and B

$$Period = \frac{2\pi}{b} = \frac{2\pi}{2} = \pi$$
$$Period = \frac{2\pi}{\frac{1}{3}} = 2\pi \div \frac{1}{3} = 2\pi \times 3 = 6\pi$$

Parts C, D and E

$$Period = \frac{2\pi}{b} = \frac{2\pi}{\pi} = 2$$
$$Period = \frac{2\pi}{e}$$
$$Period = \frac{2\pi}{\frac{1}{\pi}} = 2\pi^2$$

G. Horizontal Shift: *c*

2.37: Horizontal Shift

The graph of $f(x+k)$ is shifted horizontally:

- left by k units, $k > 0$
- right by k units, $k < 0$

$$f(x) = \sin(x-1)$$
$$f(1) = \sin(1-1) = \sin 0 = 0$$

2.38: Shifting Trig Graphs

The graph of $y = \sin(x+c)$ is shifted

- Left by c units for $c > 0$
- Right by c units for $c < 0$

Example 2.39

Identify the horizontal shift in the following, and then draw the graph:

A. $y = \sin x$
B. $y = \sin(x+\pi)$
C. $y = \sin\left(x - \frac{\pi}{2}\right)$
D. $y = \sin(x+2\pi)$

$$No\ horizontal\ shift$$
$$Shifted\ left\ by\ \pi\ Unit$$
$$Shifted\ right\ by\ \frac{\pi}{2}\ units$$
$$Shifted\ Left\ by\ 2\pi\ Units \Rightarrow No\ shift\ (because\ period\ is\ 2\pi)$$

2.40: Combining Horizontal Shift with Horizontal Scaling

Horizontal shift is to be applied *after* the horizontal scaling.

$$y = \sin\big(b(x+c)\big) = \sin(bx+bc)$$

Horizontal shift is to be applied *before* the horizontal scaling:

$$y = \sin(bx + c)$$

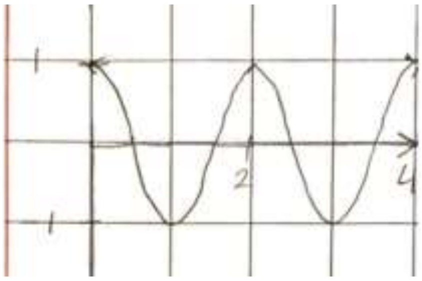
Example 2.41

$$y = \sin\left(\pi x + \frac{\pi}{2}\right)$$

$$y = \sin\left(\pi\left(x + \frac{1}{2}\right)\right)$$

$$Horizontal\ scale: Make\ the\ period\ 2$$

$$Horizontal\ Shift: \frac{1}{2} to\ the\ left$$



Example 2.42

$$y = \underbrace{\sin\left(\frac{1}{2}x + \pi\right)}_{Part\ A} = \underbrace{\sin\left(\frac{1}{2}(x + 2\pi)\right)}_{Part\ B}$$

Part A

$$Vertical\ Dilation\ Factor = 1$$

$$Max = a + d = 1 + 0 = 1$$

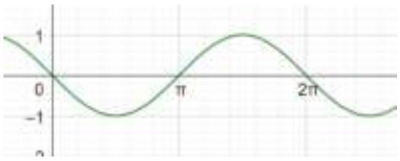
$$Min = -a + d = -1 + 0 = -1$$

$$Period = \frac{2\pi}{\frac{1}{2}} = 4\pi, Horizontal\ Scaling = 2$$

$$Horizontal\ Shift = \pi\ to\ the\ Left$$

$$Midline = d = 0$$

You need to apply the transformations in the reverse order of operations:
Shift $y = \sin x$ to the left by π units:



Stretch it horizontally by a factor of 2:

Part B

$$Max = a + d = 1 + 0 = 1$$

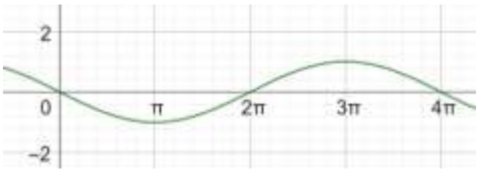
$$Min = -a + d = -1 + 0 = -1$$

$$Period = \frac{2\pi}{\frac{1}{2}} = 4\pi$$
$$Horizontal\ Shift = 2\pi$$
$$Midline = d = 0$$

You still need to apply the transformations in the reverse order of operations.

Stretch it horizontally by a factor of 2:

Shift the above graph to the left by 2π units:

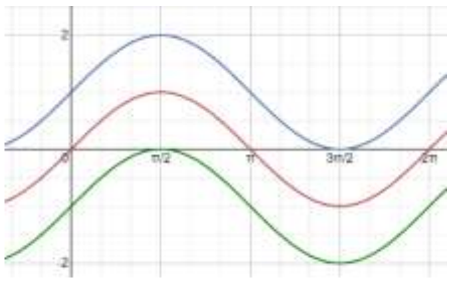


H. Vertical Shift: *d*

2.43: Vertical Translations

For $k > 0$:

$$y = f(x) + k \rightarrow Moved\ up\ k\ units$$
$$y = f(x) - k \rightarrow Moved\ down\ k\ units$$



Example 2.44

The following have been graphed in the diagram. Identify the graph and state its color.

A. $y = \sin x$
B. $y = \sin x + 1$
C. $y = \sin x - 1$

$$A: Red\ Graph$$
$$B: Blue\ Graph$$
$$C: Green\ Graph$$

Example 2.45

Identify the max and the min for the following graphs:

A. $y = \sin x$
B. $y = \sin x + 2$
C. $y = \sin x - \frac{1}{2}$
D. $y = \sin x + \pi$

$$Max = 1, \qquad Min = -1$$

$$Max = 1 + 2 = 3, \qquad Min = -1 + 2 = 1$$
$$Max = 1 - \frac{1}{2} = \frac{1}{2}, \qquad Min = -1 - \frac{1}{2} = -\frac{3}{2}$$
$$Max = 1 + \pi, \qquad Min = -1 + \pi$$

I. Max and Min

2.46: Max and Min

The max and min for $y = a\sin(bx + c) + d$ is:

$$Max = a + d$$
$$Min = -a + d$$

2.47: Midline

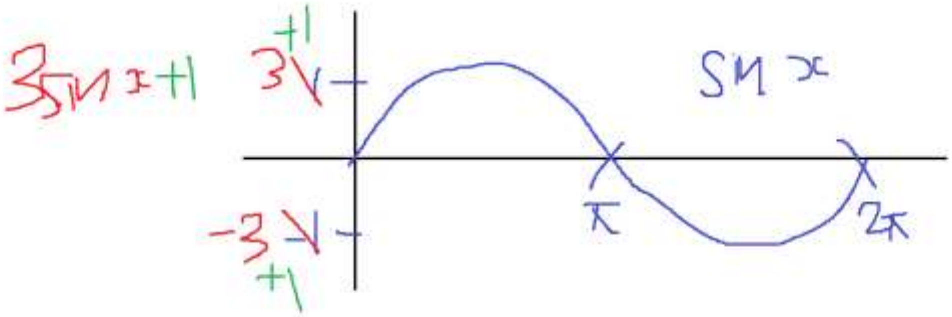
$$Midline = \frac{Max + Min}{2} = \frac{(a+d) + (-a+d)}{2} = \frac{2d}{d} = d$$

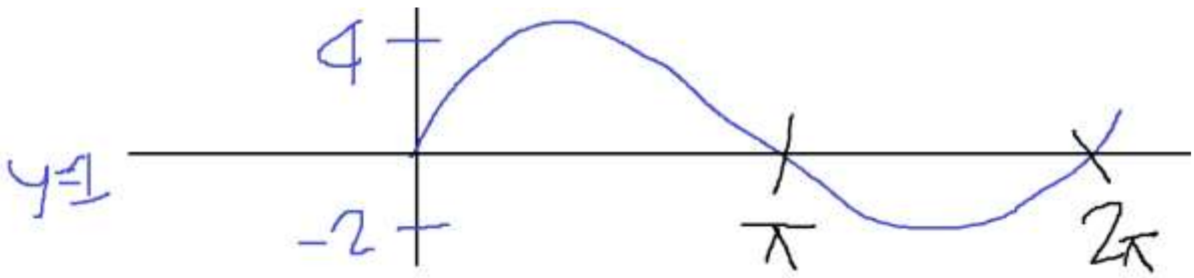
Example 2.48

Find the max, min, and midline for the following functions, and then graph them:

$$y = 3\sin(x) + 1$$

$$a = 3, d = 1$$
$$Max = a + d = 4$$
$$Min = -a + d = -3 + 1 = -2$$
$$Midline = d = 1$$





Example 2.49

Find the max, min, and midline for the following functions, and then graph them:

A. $y = \frac{1}{3}\sin(x) + 0.4$

B. $y = \pi \sin(x) + e$

Part A

$$a = \frac{1}{3}, d = 0.4 = \frac{4}{10} = \frac{2}{5}$$
$$Max = a + d = \frac{1}{3} + \frac{2}{5} = \frac{5}{15} + \frac{6}{15} = \frac{11}{15}$$
$$Min = -a + d = -\frac{1}{3} + \frac{2}{5} = -\frac{5}{15} + \frac{6}{15} = \frac{1}{15}$$
$$Midline = d = 0.4$$

Part B

$$a = \pi, d = e$$
$$Max = a + d = \pi + e$$
$$Min = -a + d = -\pi + e$$
$$Midline = d = e$$

J. Reflection

2.50: Midline

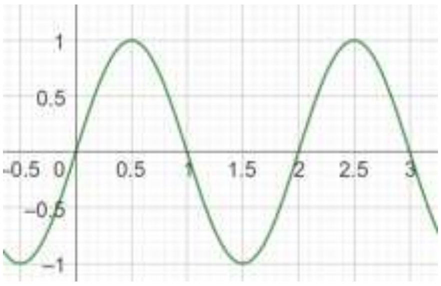
- $-f(x)$ is a reflection across the x axis
- $f(-x)$ is a reflection across the y axis

Example 2.51

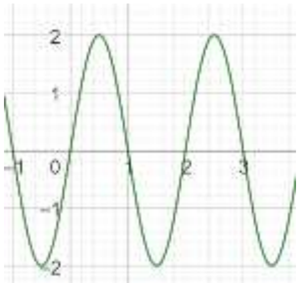
$$y = -2\sin(\pi x + \pi) + 1$$

$$y = -2\sin\big(\pi(x+1)\big) + 1$$
$$Period = \frac{2\pi}{b} = \frac{2\pi}{\pi} = 2$$

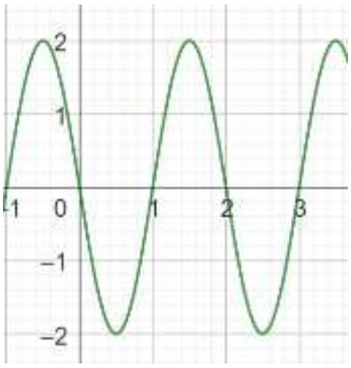
Draw $y = \sin(\pi x)$



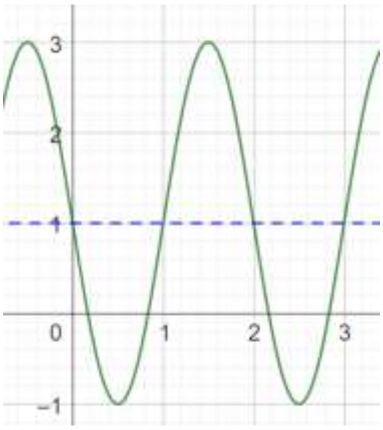
$Amplitude = 2$. Hence, draw $y = 2\sin(\pi x)$ by changing the max to 2, and the min to -2, while keeping the midline the same:

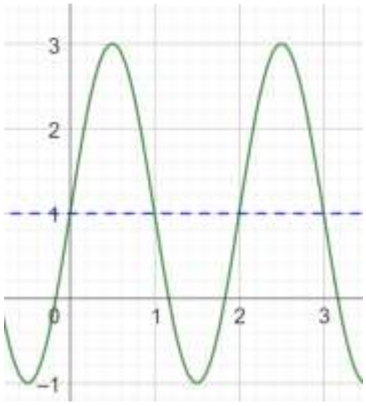


Since the graph has a reflection apply the reflection across the x axis. Draw $y = -2\sin(\pi x)$



Shift the graph up by 1 unit to incorporate the vertical shift. Draw $y = -2\sin(\pi x) + 1$ ($left\ diagram$).
Shift the graph left by 1 unit to incorporate the horizontal shift ($right\ diagram$).





K. Comparing Transformations

Example 2.52

Vayuna drew the same graph for the two functions below

$$f(x) = \sin(x) + \pi, \qquad g(x) = \sin(x + \pi)$$

A. Was she correct? If she wasn't, explain the difference between the two graphs.
B. Draw the graphs of the two functions.

$f(x)$ is a vertical translation. It moves $y = \sin x$ up by π units
$g(x)$ is a horizontal translation. It moves $y = \sin x$ left by π units

L. Combining Transformations

Example 2.53

$$y = -2\sin x + 3$$

Example 2.54

Graph

$$y = 3\cos 2x - 1$$

$$Max = 3 - 1 = 2$$
$$Min = -3 - 1 = -4$$
$$Period = \frac{360°}{b} = \frac{360°}{2} = 180°$$
$$Horizontal\ Shift = 0$$
$$Midline = d = -1$$

Example 2.55

Graph:

A. $y = \sin(3x) + 2$
B. $y = 2\sin\left(\frac{1}{2}x\right)$

Part A

$$Max = a + d = 1 + 2 = 3$$
$$Min = -a + d = -1 + 2 = 1$$
$$Period = \frac{2\pi}{b} = \frac{2\pi}{3}, \qquad Horizontal\ Scaling\ Factor = \frac{1}{3}$$
$$Horizontal\ Shift = 0$$
$$Midline = d = 2$$

Part B

$$Vertical\ Dilation\ Factor = 2$$
$$Max = a + d = 2 + 0 = 2$$
$$Min = -a + d = -2 + 0 = -2$$
$$Period = \frac{2\pi}{\frac{1}{2}} = 4\pi, \qquad Horizontal\ Scaling\ Factor = 2$$
$$Horizontal\ Shift = 0$$
$$Midline = d = 0$$

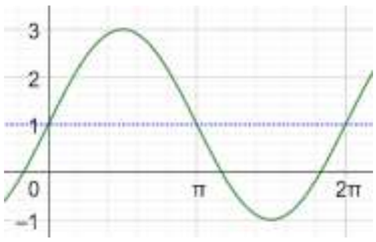
Example 2.56

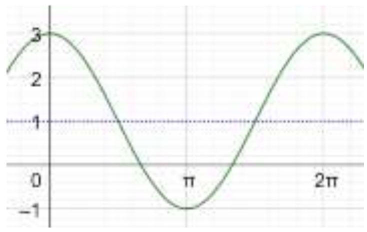
$$y = \underbrace{2\sin\left(2x + \frac{\pi}{2}\right) + 1}_{Part\ A} = \underbrace{2\sin\left(2\left(x + \frac{\pi}{4}\right)\right) + 1}_{Part\ B}$$

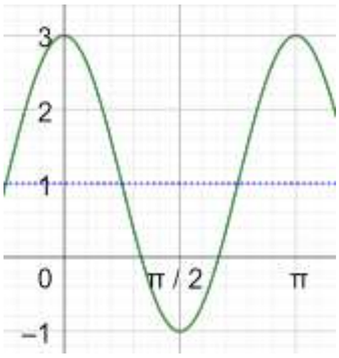
Part A

$$Vertical\ Dilation\ Factor = 2$$
$$Vertical\ Shift = 1$$
$$Max = a + d = 2 + 1 = 3$$
$$Min = -a + d = -2 + 1 = -1$$
$$Period = \frac{2\pi}{2} = \pi, Horizontal\ Scaling\ Factor = \frac{1}{2}$$
$$Horizontal\ Shift = \frac{\pi}{2}\ to\ the\ left$$
$$Midline = d = 1$$

The key is to be careful about the order in which you apply the horizontal shift and the horizontal scale. You need to apply the transformations in the reverse order of operations.







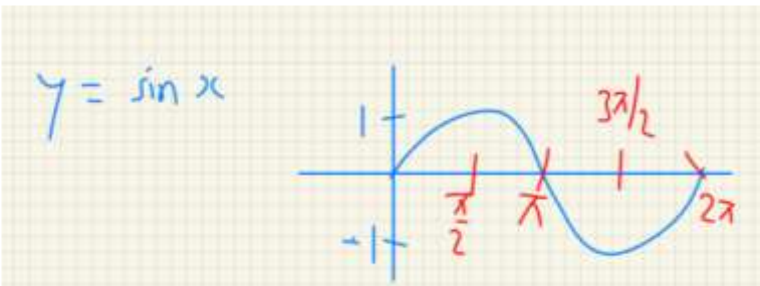
- Graph I: $y = 2\sin(x) + 1$
- Graph II: Shift Graph I horizontally to the left by $\frac{\pi}{2}$ units to get $= 2\sin\left(x + \frac{\pi}{2}\right) + 1$
- Graph III: Scale Graph II horizontally by a factor of $\frac{1}{2}$ to make its period π from 2π.

Part B

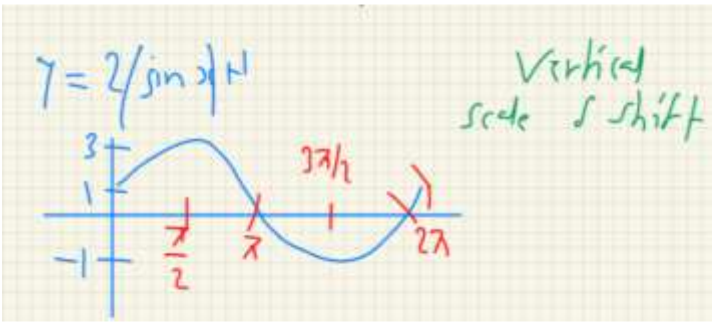
$$y = \underbrace{2\sin\left(2\left(x + \frac{\pi}{4}\right)\right) + 1}_{Part\ B}$$
$$Vertical\ Dilation\ Factor = 2$$
$$Vertical\ Shift = 1$$
$$Max = a + d = 2 + 1 = 3$$
$$Min = -a + d = -2 + 1 = -1$$
$$Period = \frac{2\pi}{2} = \pi, Horizontal\ Scaling\ Factor = \frac{1}{2}$$
$$Horizontal\ Shift = \frac{\pi}{4}\ to\ the\ left$$
$$Midline = d = 1$$

Apply the transformation in reverse of order of operations: horizontal shift after horizontal scale.

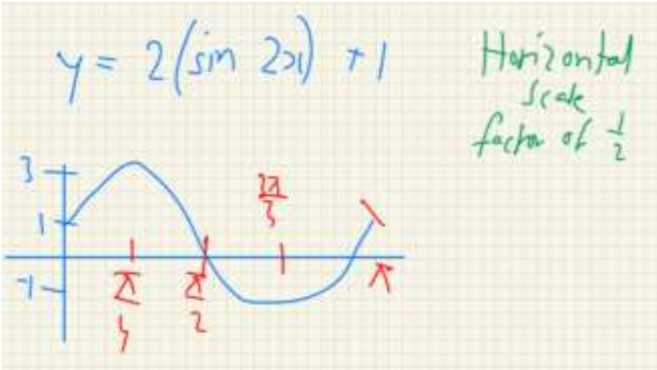
- $y = 2\sin x + 1$



- $y = 2\sin x + 1$

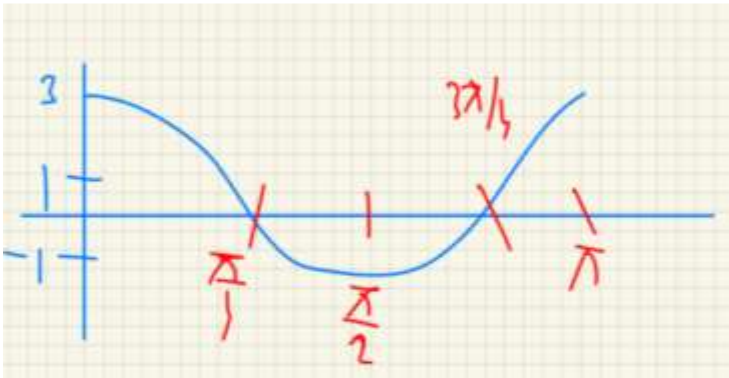


➢ $y = 2(\sin 2x) + 1$



➢ $y = 2\sin\left(2\left(x + \frac{\pi}{4}\right)\right) + 1$

Horizontal shift to the left by $\frac{\pi}{4}$



Example 2.57

Find the range of

A. $2 - 7\sin(45x)$, where x is in degrees

$$Midline = 2$$
$$Amplitude = 7$$
$$Max = 2 + 7 = 9$$
$$Min = 2 - 7 = -5$$
$$Range \in [-5,9]$$

Example 2.58

Give an equation with Max is (4,2), Min is (8,-4)

$$y = a\sin(bx + c) + d$$
$$Amplitude = \frac{2 - (-4)}{2} = 3 = a$$
$$Midline = Vertical\ Shift = \frac{-4 + 2}{2} = -1 = d$$
$$Period = 16$$

$$Phase\ Shift = 2$$
$$y = 3\sin\left(\frac{\pi}{4}(x - 2)\right) - 1$$

Example 2.59

Give an equation with Max is (7,6), Min is (12,-2)
Input in degree

$$y = a\sin(bx + c) + d$$
$$Amplitude = \frac{6 - (-2)}{2} = 4 = a$$
$$Midline = Vertical\ Shift = \frac{6 - 2}{2} = 2 = d$$

$$Max\ to\ Min = 12 - 7 = 5$$
$$Period = 2(5) = 10$$

$$\frac{2\pi}{b} = 10 \Rightarrow b = \frac{2\pi}{10} = \frac{\pi}{5}$$

$$Phase\ Shift = 2$$
$$y = 4\sin\left(\frac{\pi}{5}(x - 4.5)\right) + 2$$

Conversion factor from radians to degrees is $\frac{180}{\pi}$:

$$y = 4\sin\left(\frac{180}{\pi} \times \frac{\pi}{5}(x - 4.5)\right) + 2$$
$$y = 4\sin\big(36(x - 4.5)\big) + 2$$

2.3 Modelling

A. Modelling

Trigonometric equations exhibit periodicity. Many natural phenomena also exhibit periodicity. Hence, trigonometric functions are useful while modelling

- Biology
  - ✓ Population of certain animals over a period of time
- Weather:
  - ✓ Temperature
  - ✓ Tides
  - ✓ Rainfall
- Circular Motion: When an object moves along the circumference of a circle, or it rotates, it is called

circular motion. An important case of circular motion is uniform circular motion, where the angular rate of rotation is constant. Examples of circular motion include:

- ✓ The movement of a Ferris Wheel
- ✓ Rotation of the earth about its axis

- ➢ Simple Harmonic Motion: The motion of a frictionless spring which has a weight attached to it, and then moved from its equilibrium position follows Hooke's Law. If you plot the vertical position on the y axis, and time on the x axis, you get a sine curve.
- ➢ Waves can be modelled using trigonometric functions. Examples of waves include:
  - ✓ Waves in the Ocean
  - ✓ Sound Waves
  - ✓ Electromagnetic Waves

B. Ferris Wheels

Example 2.60

A Ferris wheel has a diameter of 11 $meters$. Its center is 7.5 $meters$ above the ground. Vyas sits at the bottom of the Ferris wheel at time $t = 0$. The Ferris Wheel completes one revolution in 5 minutes. Model this using a suitable trigonometric function of the form

$$y = a\sin(bx + c) + d$$

A. What is the amplitude a?
B. What is the midline?
C. What is the vertical shift d?
D. What is the period? Hence, what is the value of b?
E. What is the horizontal shift c?
F. What is the function that models Vyas' height at time t?

$$Amplitude = a = \frac{Max - Min}{2} = \frac{13 - 2}{2} = \frac{11}{2} = 5.5$$

Note that the amplitude is simply the radius of the Ferris wheel.

$$Midline = \frac{Max + Min}{2} = \frac{13 + 2}{2} = \frac{15}{2} = 7.5$$

Note that the midline is simply the height of the center of the Ferris wheel.

The vertical shift matches the midline:

$$Vertical\ Shift = d = 7.5$$

$$Period = 5\ min$$

Substitute the time take for the Ferris wheel to go around once (5 min) in the equation for Period, which is $Period = \frac{2\pi}{b}$

$$5 = \frac{2\pi}{b} \Rightarrow b = \frac{2\pi}{5}$$

Method I: Horizontal shift first

At time $t = 0$, the height is $h = 2$, which means the graph is at a minimum.

$$Horizontal\ Shift = \frac{\pi}{2}\ units\ to\ the\ right$$

Substitute $a = 5.5, d = 7.5, b = \frac{2\pi}{5}$ in $y = a\sin(bx + c) + d$

$$y = 5.5\sin\left(\frac{2\pi}{5}t - \frac{\pi}{2}\right) + 7.5$$

Method II: Horizontal scaling first

The graph needs to shift right by $\frac{1}{4}$ of a cycle to have the minimum as the starting point (instead of the midline as the starting point).

Hence, we need to shift right by

$$\frac{5}{4} = 1.25$$

$$y = 5.5\sin\left(\frac{2\pi}{5}(t - 1.25)\right) + 7.5$$

Example 2.61

You are on a Ferris wheel of diameter 30 meters. It makes a complete revolution every 90 seconds.

A. If you get on at the bottom, write a model for your vertical position assuming the wheel stands on a 2 meters high platform.
B. Calculate the height of your seat after 15 seconds.
C. When are you at a height of 28 meters? Write your answer in exact form using the cos inverse function only.

Part A

$$Amplitude = a = \frac{Max - Min}{2} = \frac{30}{2} = 15$$
$$Vertical\ Shift = Midline = d = \ Amplitude + Min = 2 + 15 = 17$$

$$Period = 90 = \frac{2\pi}{b} \Rightarrow b = \frac{2\pi}{90} = \frac{\pi}{45}$$
$$Horizontal\ Shift = c = \frac{\pi}{2}\ units\ to\ the\ right$$

$$y = 15\sin\left(\frac{\pi}{45}t - \frac{\pi}{2}\right) + 17$$

Part B

Substitute $\sin\left(\frac{\pi}{45}\cdot 15 - \frac{\pi}{2}\right) = \sin\left(\frac{\pi}{3} - \frac{\pi}{2}\right) = \sin\left(-\frac{\pi}{6}\right) = -\sin\frac{\pi}{6} = -\frac{1}{2}$ in the expression for y:

$$= 15\left(-\frac{1}{2}\right) + 17 = -7.5 + 17 = 9.5$$

Part C

$$15\sin\left(\frac{\pi}{45}t - \frac{\pi}{2}\right) + 17 = 28$$
$$15\sin\left(\frac{\pi}{45}t - \frac{\pi}{2}\right) = 11$$
$$\sin\left(\frac{\pi}{45}t - \frac{\pi}{2}\right) = \frac{11}{15}$$
$$-\cos\left(\frac{\pi}{45}t\right) = \frac{11}{15}$$
$$\cos\left(\frac{\pi}{45}t\right) = -\frac{11}{15}$$

$$\frac{\pi t}{45} = \cos^{-1}\left(-\frac{11}{15}\right)$$
$$t = \frac{45}{\pi}\cos^{-1}\left(-\frac{11}{15}\right)$$

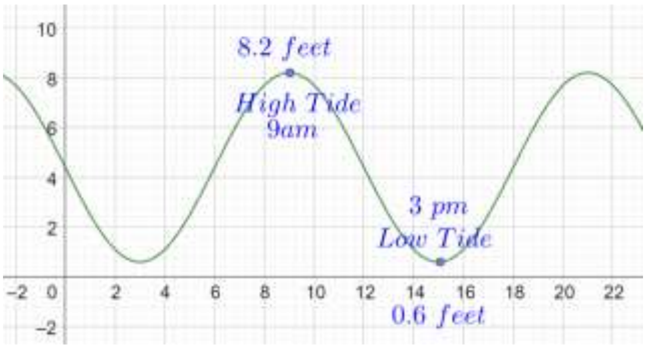
C. Tides

Example 2.62

At a certain location, high tide occurred at 9 am, with a water height of 8.2 feet. The next low tide occurred at 3 pm, with a water height of 0.6 feet. Write an equation that represents this situation starting from $time\ t = 0$, which is midnight.

Let t be the number of hours after midnight. We want an equation of the form:

$$y = a \sin b(t - c) + d$$
$$Amplitude = a = \frac{Max - Min}{2} = \frac{8.2 - 0.6}{2} = \frac{7.6}{2} = 3.8$$
$$Midline = d = \frac{Max + Min}{2} = \frac{8.2 + 0.6}{2} = \frac{8.8}{2} = 4.4$$



$$High\ Tide\ to\ Low\ Tide = 6\ Hours$$
$$High\ Tide\ to\ High\ Tide = 12\ Hours$$

$$Period = \frac{2\pi}{b} = 12 \Rightarrow b = \frac{\pi}{6}$$

So far, we have:

$$y = 3.8 \sin\left[\frac{\pi}{6}(t - 6)\right] + 4.4$$

(Calc) Example 2.63

At 2:00 pm on August 2^{nd}, at high tide, you find that the depth of the water is 1.5 m. At 7: 30 pm, at low tide, the depth of the water is 1.1 m. Assume that the depth varies sinusoidally with time.

A. Find an equation expressing depth as a function of time that has elapsed since 12: 00 am, August 2.
B. Predict the depth of the water at 5: 00 pm on August 3.
C. When does the first low tide occur on August 3?
D. Find the earliest time on August 3 that the water depth will be at 1.27 m? Answer to the nearest second.

Part A

$$d = Midline = \frac{Max + Min}{2} = \frac{1.5 + 1.1}{2} = 1.3$$
$$Period = 2(7.30\ pm - 2.00\ pm) = 2(5.5) = 11\ hours$$
$$\frac{2\pi}{b} = 11 \Rightarrow b = \frac{2\pi}{11}$$
$$a = Amplitude = \frac{Max - Min}{2} = \frac{1.5 - 1.1}{2} = \frac{0.4}{2} = 0.2$$

High Tide is at:

$$2.00\ pm - 11\ hours = 3\ am$$

Consider $f(t) = \sin\left(\frac{2\pi}{11}t\right)$. The max, which is high tide, occurs in the above equation at

$$\frac{2\pi}{11}t = \frac{\pi}{2} \Rightarrow t = \frac{\pi}{2} \cdot \frac{11}{2\pi} = \frac{11}{4} = 2.75\ hours$$

Hence, we shift it right by 0.25 hours to get:

$$f(t) = \sin\left(\frac{2\pi}{11}(t - 0.25)\right)$$

And add the amplitude and midline to get the final equation:

$$f(t) = 0.2\sin\left(\frac{2\pi}{11}(t - 0.25)\right) + 1.3$$

Part B

5: 00 pm on August 3 is

$$41\ hours\ after\ 12:00\ am\ on\ Aug\ 2 \Rightarrow t = 41$$

We need to find:

$$f(41) = 1.11$$

Part C

$$7.30\ pm, Aug\ 2 + 11\ Hours = 6.30\ am, Aug\ 3$$

Part D

$$f(t) = 1.27$$

D. Ocean Waves

Example 2.64

You observe a buoy in the ocean bobbing up and down a total distance of 8 feet. It starts at the bottom of a wave and completes four cycles in one minute. Write a model that describes its vertical position in terms of t minutes.

$$Amplitude = \frac{Max - Min}{2} = \frac{8}{2} = 4$$

$$Midline = d = 0$$

$$Period = \frac{2\pi}{b} = \frac{1}{4} \Rightarrow b = 8\pi$$

The graph of sin x needs to be shifted one fourth of a cycle to the right, which is:

$$-\frac{1}{4} \times \frac{1}{4} = -\frac{1}{16}$$

Hence, the final answer is:

$$y = 4\sin\left(8\pi\left(t - \frac{1}{16}\right)\right)$$

E. Biology

Example 2.65

Naturalists find that populations of some kinds of predatory animals vary periodically with time. Assume that the population of foxes in a certain forest varies sinusoidally with time. Records started being kept at time $t = 0$ year. A minimum number of 200 foxes appeared at $t = 2.9$ years. The next maximum, 800 foxes, occurred at $t = 5.1$ years.

A. Sketch the graph of this sinusoid.
B. Find a particular equation expressing the number of foxes as a function of time.
C. Predict the fox population when $t = 7, 8, 9,$ and 10 years.
D. Suppose foxes are declared a vulnerable species when their population drops below 300. Between what two non-negative values of t did the foxes become vulnerable?

F. Sound

Example 2.66

The hum you hear on some radios when they are not tuned to a station is a sound wave of 60 cycles per second.

A. Is 60 cycles per second the period or is it the frequency? If it is the period, find the frequency. If it is the frequency, find the period.
B. The wavelength of a sound wave is defined as the distance the wave travels in a time interval equal to one period. If sound travels at 1100 ft/s, find the wavelength of the 60-cycle-per-second hum.
C. The lowest musical note the human ear can hear is about 16 cycles per second. In order to play such a note, a pipe on an organ must be exactly half as long as the wavelength. What length organ pipe would be needed to generate a 16-cycle-per-second note?

Part A

$$Frequency = f = 60\frac{cycles}{second}$$
$$Period = \frac{1}{f} = \frac{1}{60}\ Seconds$$

Part B

$$Speed\ of\ Sound = 1100\frac{ft}{s}$$

The wave length is the distance travelled in one period:

$$Wavelength = Distance = \underbrace{1100\frac{ft}{sec}}_{Speed} \times \underbrace{\frac{1}{60}}_{Time} sec = \frac{1100}{60} = \frac{110}{6} = \frac{55}{3} = 18\frac{1}{3}\ ft$$

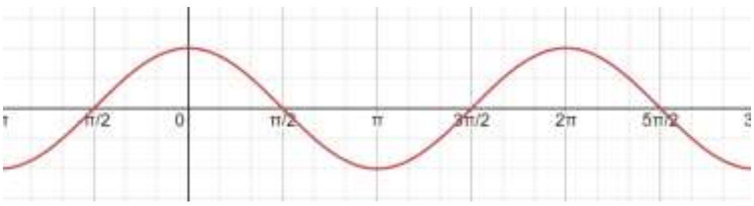
Part C

$$Pipe\ Length = Wavelength \times \frac{1}{2} = \frac{1100}{16} \times \frac{1}{2} = 34\frac{3}{8}$$

2.4 Graphs: Transformations

A. Basics

Example 2.67



A. The graph of $\cos x$ is the graph of $\sin x$ shifted by a units. Write the graph in this form.
B. What are the roots of $y = \cos x$?
C. What is the period of $y = \cos x$?
D. What is the amplitude?
E. What is the midline?
F. What is the domain of $y = \cos x$?
G. What is the range of $y = \cos x$?

Part A

$$y = \sin\left(x + \frac{\pi}{2}\right)$$

Part B

$$x = \left\{ \dots, -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots \right\}$$
$$x = \frac{n\pi}{2}, \qquad n\ is\ odd\ integer$$

Part C

Shifting the graph of $y = \sin x$ to the left (as we did in Part A) does not change its period. Hence,
$$Period = 2\pi$$

Part D

$$Amplitude = 1$$

Part E

$$y = 0$$

Part F

$$Domain = (-\infty, \infty) = \mathbb{R}$$

Part G

$$Range = [-1,1]$$

B. Graphing

Example 2.68

Identify the transformation applied to $y = \cos x$ to obtain the following functions.

A. $y = \cos x + 2$
B. $y = \cos(x + 2)$
C. $y = 2\cos x$
D. $y = \cos\left(\frac{x}{2}\right)$

Example 2.69

Identify the midline, amplitude, period and range for the function below:
$$y = 3\cos\left(\frac{x}{\pi} + 1\right) - 2$$

$$Midline: y = -2$$
$$Amplitude = 3$$
$$Period = \frac{2\pi}{\frac{1}{\pi}} = 2\pi^2$$
$$Range = [1, -5]$$

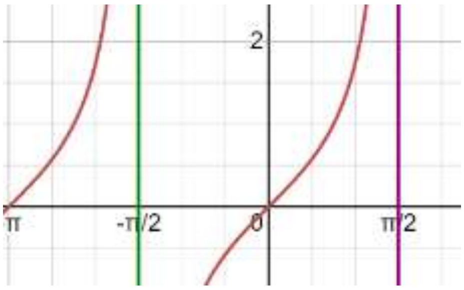
C. tan x

2.70: Domain

The domain of a function $y = f(x)$ is the set of acceptable x values for the function.

Two standard types of expressions which restrict the domain of an expression are:

- Fractions: The denominator must be non-zero, since you cannot divide by zero.
- Square Roots: In the real number system, the expression inside a square root must be positive.



Example 2.71

What is the domain of $f(x) = \tan x$?

$$\tan x = \frac{\sin x}{\cos x}$$

The above is not defined when $\cos x = 0$, which is precisely when:

$$x = \frac{\pi}{2} + k\pi, \qquad k \in \mathbb{Z}$$

2.72: Domain of $\tan x$

$$Domain = \mathbb{R} - \left\{\frac{\pi}{2} + k\pi, \qquad k \in \mathbb{Z}\right\}$$

The domain of $\tan x$ is all real numbers except the values when $\cos x = 0$, or the roots of $y = \cos x$.

Example 2.73

What is the domain of $f(x) = \frac{\tan x}{\sec x}$?

$$\frac{\tan x}{\sec x} = \frac{\frac{\sin x}{\cos x}}{\frac{1}{\cos x}}$$

Since we have $\cos x$ in the denominator in two places, we cannot have

$$\cos x = 0 \Rightarrow x = \left\{\frac{\pi}{2} \pm k\pi, \qquad k \in \mathbb{N}\right\}$$

Example 2.74

What is the domain of $f(x) = \frac{\tan x}{\sin x}$?

$$\frac{\tan x}{\sin x} = \frac{\frac{\sin x}{\cos x}}{\sin x}$$

$$\cos x = 0 \Rightarrow x = \left\{\frac{\pi}{2} \pm k\pi, \qquad k \in \mathbb{N}\right\} = A$$

$$\sin x = 0 \Rightarrow x = \{k\pi, \qquad k \in \mathbb{Z}\} = B$$

$$Domain = \mathbb{R} - (A \cup B)$$

Example 2.75

What is the range of $y = \tan x$

$$Range = \mathbb{R}$$

Example 2.76

What are the roots of $y = \tan x$?

$$x = n\pi, \qquad n \in \mathbb{Z}$$

Example 2.77

The period of $y = \tan\left(\frac{x}{a\pi}\right)$ is the same as the period of $y = \sin(xb\pi)$. The value of ab is equal to a constant. Determine the constant.

$$\frac{\pi}{\frac{1}{a\pi}} = \frac{2\pi}{b\pi} \Rightarrow \pi \times a\pi = \frac{2}{b} \Rightarrow ab = \frac{2}{\pi^2}$$

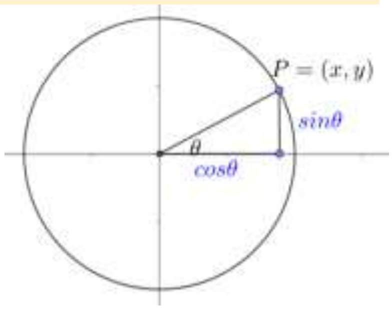
D. Sums of Functions

Example 2.78

Period

Maximum and Minimum

Range



79 Examples